



Rethinking currency factors: The case for mean-variance optimisation[☆]

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HIGHLIGHTS

- Bayesian mean-variance optimisation significantly improves currency carry, momentum, and value strategies.
- The outperformance is driven by improved cross-sectional allocation rather than enhanced time-series predictability.
- Optimised factors exploit negative cross-covariances and dispersion in expected currency returns.
- The MV-optimised factors provide a novel asset pricing framework.

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ABSTRACT

We show that a Bayesian mean-variance (MV) optimisation can substantially enhance the performance of established currency factor strategies such as carry, value, and momentum. We find that the improved performance is due to the stronger cross-sectional predictability of the optimised strategies, instead of the time series predictability. International diversification plays an important role in the outperformance of the MV optimisation approach. Our asset pricing tests suggest that the MV-optimised factors subsume the abnormal return of the naïve currency factors.

1. Introduction

Empirical studies in equity markets have shown that sophisticated portfolio optimisation approaches often fail to outperform naïve diversification due to estimation error and the high dimensionality of the asset universe (DeMiguel et al., 2009b; Hsu et al., 2018; Yuan and Zhou, 2024). In contrast, currency markets exhibit features that make portfolio optimisation more promising. First, the cross-section of investable currencies is relatively small, as FX trading is heavily concentrated in a limited set of major currencies, particularly the G10.¹ Empirical currency studies, therefore, typically focus on a small cross-section or extend the sample to up

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¹ According to the 2025 BIS Triennial Survey, the US Dollar was on one side of 89.2% of all FX trades, followed by the Euro (28.9%), Japanese Yen (16.8%), UK Pound Sterling (10.2%), Swiss Franc (6.4%), and the Australian and Canadian Dollars (around 6% each).

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to 48 currencies as in Menkhoff et al. (2012b), which substantially reduces covariance estimation error relative to equity markets (Ackermann et al., 2017). Second, expected returns are concentrated in a small number of well-established factors such as carry, momentum, and value, rather than diffuse firm-level characteristics (Baz et al., 2001; Della Corte et al., 2009; Jamali et al., 2023). Consistent with this view, Chernov et al. (2023) show that portfolio optimisation is both feasible and economically meaningful in currency markets. This suggests that portfolio optimisation can provide a useful framework for constructing currency factors and studying risk premia.

In this paper, we provide a link between currency factors and portfolio optimisation by proposing a Bayesian mean-variance optimiser on extant currency factors. Specifically, we optimise three of the most extensively documented currency factor portfolios, namely carry, momentum and value. We term these portfolios as mean-variance (MV) optimised factors. The MV-optimised factors are constructed based on information conditional on expected currency returns, the covariance matrix and currency characteristics. We find that the MV-optimised factors significantly outperform both their corresponding plain currency factors and the mean-variance optimised portfolio, unconditional on currency characteristics.

Through a performance attribution analysis, we investigate whether the outperformance of MV-optimised factors is due to their enhanced time series or cross-sectional predictability. Generalising the decomposition model of Lo and MacKinlay (1990) and Lewellen (2002) to the currency trading strategies, we find that the diversification-benefit component generated by cross-covariances is much stronger for optimised factors compared to naïve factors. This means that optimised currency factors can better capture the cross-sectional variation of currency returns. This makes sense as in a plain currency factor, once the long and short portfolios are determined, each asset is assigned an equal weight without consideration of their comovement. In contrast, our MV-optimised factors incorporate covariance matrix information into the long and short portfolios, leading to better cross-sectional predictability.

We find that international diversification plays a pivotal role in explaining the enhanced performance of the MV-optimised factors, as their performance is more pronounced in periods when the investable currencies are more diversified between developed and emerging economies. More specifically, to rationalise the high excess returns of the MV-optimised factors, we highlight three reasons: 1) the low correlation between global currencies, 2) the low level of comovement across the emerging currencies relative to developed currencies, and 3) the dispersion in distributions of currency abnormal returns. More recent studies on currency market optimisation, such as Ackermann et al. (2017), Maurer et al. (2022), Maurer et al. (2023) and Chernov et al. (2023), mainly focus on using a small cross-section of currencies.² However, we provide new evidence that highlights the importance of diversification by adopting a relatively larger cross-section of 48 currencies as in Menkhoff et al. (2012a) and Menkhoff et al. (2012b). Similar to the dynamic in equity markets, where a stock with a negative beta is deemed a valuable asset in a portfolio, a currency with a lower correlation to the main currencies is also deemed to be valuable. Furthermore, to address concerns about liquidity and market depth, we also conduct MV-optimised factors based on a smaller sample as in Ackermann et al. (2017), mainly consisting of developed and large emerging economies. The results indicate that even within this more liquid subsample, the MV-optimised factors continue to deliver statistically significant outperformance, reinforcing the robustness of our optimisation framework.

DeMiguel et al. (2009b) find that the optimal portfolios consistently fail to outperform the naïve portfolio when assuming every asset's CAPM α is equal to zero. However, even though empirically the mean of α is quite close to zero (Jarrow, 2010), the assumption of a precisely zero CAPM is not realistic in the real world. Motivated by this, Platanakis et al. (2021) allow for a zero mean distribution of CAPM α and a standard deviation of 30 basis points. Their results show that the MV portfolios significantly outperform the 1/N rule. Our empirical finding in the currency market is consistent with Platanakis et al. (2021), where the outperformance of MV factor portfolios is positively related to the cross-sectional standard deviation of currency abnormal returns. According to our results of the *F*-test, the cross-sectional standard deviation of emerging currencies' abnormal returns significantly exceeds that of the developed sample at the 1% level, confirming the importance of international diversification.

We further explore the asset pricing implications of the MV-optimised factors and examine the relationship between these optimised factors and their naïve counterparts in currency pricing models. This analysis relates to a growing literature that studies the pricing of currency risk and the dynamics of currency portfolio returns in international financial markets, e.g., Clarida et al. (2009), Byrne et al. (2019), and Hoffmann and Studer-Suter (2017). First, our spanning tests reveal that all three MV-optimised factors fully subsume the abnormal returns of the naïve factors. Second, based on the framework of Fama and French (2015), we assess the performance of MV currency factors in multi-factor currency pricing models. Adding the MV-optimised factors into the existing currency pricing model can significantly improve the model's predictability across all four evaluation metrics. Specifically, a four-factor model comprising the Dollar Index, the MV-optimised momentum factor, and two other naïve factors outperforms all other specifications.

Our study contributes to the understanding of the role of MV optimisation in improving the performance of existing currency factors. Our research is different from studies that directly adapt optimisation for currencies without factor signals, such as Baz et al. (2001) and Ackermann et al. (2017), or that focus on improving the estimation of currency expected returns based on factor signals, such as Opie and Riddiough (2020), Maurer et al. (2022), and Chernov et al. (2023). Kroencke et al. (2014) proposed a combined signal based on carry, momentum and value, but simply averages the ranks across three signals without using additional information apart from currency characteristics. In contrast, we directly employ optimisation to the constituents of the long and short legs of currency factors to improve performance.

Finally, these MV-optimised factors provide economic value by enhancing currency factor portfolios, leading to superior performance compared to traditional currency factors. Despite their optimisation, the MV factors preserve the original portfolio structure,

² Ackermann et al. (2017) demonstrate that mean-variance portfolio optimisation beats naïve diversification in currency markets as it benefits from a small cross-section.

ensuring balance between long and short positions with no net exposure. Unlike Della Corte et al. (2009), Barroso and Santa-Clara (2015a), and Maurer et al. (2023), our strategy avoids leverage, ensuring that portfolio weights remain stable over time. Incorporating notional values may further refine MV optimisation, improving performance beyond the current framework. For further analysis of the profitability of optimised currency factors, see Fan et al. (2025).

The remainder of this paper is organised as follows. In Section 2, we describe our dataset and the main methodologies used in constructing currency factors and in mean-variance optimisation. Next, Section 3 compares the performance between the proposed MV-optimised factors and those using naïve diversification. Section 4 analyses the outperformance attribution of our proposed strategies. In Section 5, we test whether the MV-optimised factors can be employed in currency asset pricing. Section 6 provides a series of robustness checks. Finally, Section 7 concludes.

2. Data and factor portfolios

This section begins by outlining our data universe and the methodology for computing currency excess returns and transaction costs. We then proceed to describe the construction of currency factor portfolios and the proposed mean-variance optimisation process.

2.1. Data

We use a comprehensive FX market dataset proposed by Menkhoff et al. (2012a) and Menkhoff et al. (2012b), which consists of 48 currencies from the following countries or regions: Australia, Austria, Belgium, Brazil, Bulgaria, Canada, Croatia, Cyprus, Czech Republic, Denmark, Egypt, Euro area, Finland, France, Germany, Greece, Hong Kong, Hungary, Iceland, India, Indonesia, Ireland, Israel, Italy, Japan, Kuwait, Malaysia, Mexico, Netherlands, New Zealand, Norway, Philippines, Poland, Portugal, Russia, Saudi Arabia, Singapore, Slovakia, Slovenia, South Africa, South Korea, Spain, Sweden, Switzerland, Taiwan, Thailand, Ukraine, and the United Kingdom. Our sample period spans from November 1983 to December 2024, with summary statistics reported in Appendix A. We obtain spot and 1-month forward exchange rates for each currency from Barclays Bank International (BBI) and WM Reuters (WMR) via Datastream.

We retain this sample of Menkhoff et al. (2012a) and Menkhoff et al. (2012b) rather than expanding the universe to additional currencies for two reasons. First, the dataset provides the largest feasible cross-section of liquid currencies with reliable spot and forward markets while maintaining comparability with the benchmark literature. Using a large cross-section also represents a demanding environment for portfolio optimisation, as covariance estimation error increases with the number of assets. Second, several additional currencies outside this universe are subject to capital controls, exchange-rate management, or persistent deviations from covered interest parity (CIP), implying that their spot and forward rates may not fully reflect market-clearing prices. Including such currencies would introduce institutional distortions that confound the measurement of currency risk premia.

We first measure the monthly gross return. Following previous literature (e.g., Fama, 1984), we calculate the log spot rate change of currency k in month t as $\Delta s_t^k = s_t^k - s_{t-1}^k$, where s_t^k represents the log spot exchange rate at the end of the month, measured in foreign currency units per the United States (US) Dollar. When Δs_t^k is positive, currency k depreciates against the US Dollar.

Next, we follow Menkhoff et al. (2012b) and measure the monthly excess returns as:

$$rx_t^k = (i_{t-1}^k - i_{t-1}) - \Delta s_t^k, \quad (1)$$

where i_{t-1}^k is the foreign interest rate in region k in month $t-1$; and i_{t-1} is the US interest rate. Therefore, $i_{t-1}^k - i_{t-1}$ refers to the interest differential between economy k and the US. We follow Taylor (1989), Akram et al. (2008) and Menkhoff et al. (2012a,b) in estimating the interest rate differentials using forward discounts. Thus, we have the short-term interest differential $i_{t-1}^k - i_{t-1} \approx f_{t-1}^k - s_{t-1}^k$, where f_{t-1}^k is the log one-month forward exchange rate between currency k and the US dollar in month $t-1$ (foreign currency units per US Dollar). Then, we compute the monthly excess returns of currency k in month t as:

$$rx_t^k = (i_{t-1}^k - i_{t-1}) - \Delta s_t^k \approx f_{t-1}^k - s_{t-1}^k. \quad (2)$$

In practice, transaction costs can have a significant impact on the measurement of investment revenues, with Timmermann and Granger (2004) documenting that excess returns generated by exchange rate spreads can often be eliminated in the presence of transaction costs. Menkhoff et al. (2012b) estimates transaction costs using the full quoted bid-ask spread; however, Neely and Weller (2013) proposed that the full quoted bid-ask spreads tend to be larger than the effective spread rates that are actually traded. We follow Neely and Weller (2013) and compute one-way approximate transaction costs as one-third of the one-month forward bid-ask spread. This transaction cost measurement is extensively used in FX studies, e.g., Hsu et al. (2016) and Filippou et al. (2018). Therefore, the transaction-adjusted portfolio return is:

$$R_t^p = \ln\{1 + \sum_{k=1}^N [\exp(rx_t^k) - 1]w_t^k - \sum_{k=1}^N [|w_{t+1}^k - w_t^k| \times c_t^k]\}, \quad (3)$$

where N refers to the number of currencies in the portfolio, c_t^k represents the transaction cost for currency k in month t , w_t^k is the weight of currency k before the rebalancing at the end of month t , and w_{t+1}^k is the weight in the coming month. w_t^k is positive (negative) if the asset k is assigned to the long (short) portfolio.

2.2. Currency factors

We implement three widely used currency market anomalies: the carry factor of Lustig et al. (2011) and Menkhoff et al. (2012a), the currency momentum factor of Menkhoff et al. (2012b), and the currency value factor of Moskowitz et al. (2012) and Menkhoff et al. (2017).

Carry refers to the interest differential measured by the difference between a spot rate and a one-month forward rate, as defined by Lustig et al. (2011). Consequently, a carry strategy involves taking a long position in currencies with higher interest rates and a short position in currencies with lower interest rates. As the covered interest parity theorem holds at the frequency used in our study (Akram et al., 2008), we estimate the cost of carry using the forward discount, $f_t - s_t$, in line with the approach in Menkhoff et al. (2012a).³ A carry factor portfolio is then created by buying (selling) the currencies with high (low) costs of carry.

Momentum is a cross-sectional trend market anomaly that has been extensively documented in various financial markets. In a momentum strategy, as defined by Jegadeesh and Titman (1993), abnormal profits are generated by buying the securities that are past winners and selling those that are past losers. More recently, Menkhoff et al. (2012b) and Raza et al. (2014) have uncovered momentum profits in the currency market. In line with Menkhoff et al. (2012b), we adopt a momentum strategy with a three-month formation period as a proxy for the currency momentum anomaly. A momentum factor portfolio is subsequently constructed by going long (short) on the currencies with high (low) cumulative returns over the formation period.

Value is a type of mean reversion currency anomaly outlined in Asness et al. (2013) and Menkhoff et al. (2017). Its premium is sourced from the Real Exchange Rate (RER) as follows:

$$RER_t = spot_t \times \frac{P_t}{P_{f,t}}, \quad (4)$$

where $P_{f,t}$ and P_t refer to the foreign and domestic price level in month t , respectively. $\frac{P_t}{P_{f,t}}$ is also known as the Purchasing Power Parity (PPP). In line with Menkhoff et al. (2017), we estimate the currency value as the log difference in the real exchange rates over the past 60 months (5 years) as:

$$V_t = \log \left(\frac{spot_t * P_t}{P_{f,t}} \right) - \log \left(\frac{spot_{t-60} * P_{t-60}}{P_{f,t-60}} \right), \quad (5)$$

where $spot_{t-60}$ is the spot rate five years ago. In practice, we follow Asness et al. (2013) and adopt the average of spot rates between 4.5 and 5.5 years ago as the spot rate five years ago to smooth out short-term market noise and the impact of any single day's price volatility. Moreover, we adopt the U.S. domestic and foreign CPI indices as proxies for the price levels, P_t and $P_{f,t}$, respectively, to avoid base year effects (Asness et al., 2013; Menkhoff et al., 2017). The premise of the value strategy is that currencies with a weak RER relative to the U.S. dollar are considered 'cheap', or high-value, and are expected to have higher future returns. Therefore, the strategy is to go long on the high-value currencies and short on the low-value currencies.

In addition to the three currency factors for optimisation, we construct several benchmark factors for the asset pricing tests. First, we compute the dollar factor of Lustig et al. (2011) using our sample of 48 currencies. Second, we construct the economic and inflation momentum factors proposed by Dahlquist and Hasseltoft (2020). Moreover, we follow Dahlquist and Hasseltoft (2020) to obtain the VIX index, TED spread, and foreign exchange volatility from Bloomberg as market controls, and further incorporate the geopolitical risk index developed by Caldara and Iacoviello (2022) as an additional macroeconomic control variable.

2.3. Mean-variance optimisation

Next, we outline our mean-variance method to optimise the three currency factor portfolios presented above. Assuming N assets are available at time t , all assets are sorted in descending order based on currency characteristics for momentum and carry signals and in ascending order for value signals.

In a 50/50 factor portfolio, half of the available assets are allocated to the long portfolio, with the other half allocated to the short portfolio. As both the long and short portfolios have $N/2$ assets, the long/short portfolios can be two subsets of the entire asset universe, each containing $N/2$ assets.⁴ A given asset cannot be allocated into two different factor portfolios simultaneously, so the two subsets have no intersection.

We optimise the long and short legs separately for several reasons. First, the two subsets are characterised by distinct expected return functions. Heterogeneity in expected return functions across asset subsets can bias return forecasts and dilute the effectiveness of mean-variance optimisation (Pástor and Stambaugh, 2000).⁵ Second, risk exposures are structurally imbalanced across legs (Barroso and Santa-Clara, 2015b; Daniel and Moskowitz, 2016). For example, short legs in carry strategies often concentrate crash risk during episodes of market stress (Menkhoff et al., 2012a), whereas long legs tend to follow smoother return dynamics. Finally, the correlation

³ More recent literature shows that the covered interest parity is violated after the financial crisis (Du et al., 2018). Here, we follow the assumption of Barroso et al. (2022) that, for the majority of investors, opportunities for arbitrage in currency markets are absent, making forward or futures contracts the most practical instruments for implementing currency strategies.

⁴ Here, we assume N is an even number. However, if the N is odd, we can remove the currency ranked in the middle and take $(N - 1)/2$ assets in each of the long/short legs.

⁵ Evidence of such asymmetry in currency markets is provided by Fan et al. (2025).

between long and short legs is unstable and tends to deteriorate in periods of market stress. For instance, correlations within carry portfolios can invert during risk-off episodes, leading to pooled optimisation systematically misestimating total portfolio risk (Jurek, 2014).

Overall, by treating each leg independently, the optimiser can better accommodate distributional asymmetries to enhance out-of-sample stability (Della Corte et al., 2009) and enable targeted risk management, such as down-weighting crash-prone short positions using shrinkage estimators (Jorion, 1986), while preserving exposure to higher-conviction long positions. To implement this, we express the weights of the long/short portfolio (w_{long}/w_{short}) as two vectors,

$$\begin{aligned} w_{long} &= [w_1, w_2, \dots, w_{N/2}], \\ w_{short} &= [w_{N/2+1}, w_{N/2+2}, \dots, w_N], \end{aligned} \quad (6)$$

where w_N denotes the weight of a given asset. According to Goyal and Jegadeesh (2017), the asset weights in the long (short) portfolio should add up to one (minus one), so we hold $\sum_{i=1}^{N/2} w_i = 1$ and $\sum_{i=N/2+1}^N w_i = -1$. In an original factor portfolio, the weight of each asset in the long (short) portfolio is denoted by $\frac{1}{N/2}$ ($-\frac{1}{N/2}$).

To determine the optimal w_{long} and w_{short} , under the mean-variance framework of Markowitz (1952), it is assumed that investors choose the asset weights vector, w_t , that maximises their expected utility:

$$\max_{w_t} \quad w_t' \mu_t - \frac{\gamma}{2} w_t' \Sigma_t w_t, \quad (7)$$

where γ is their level of relative risk aversion. The selection of γ becomes less consequential in our method, as the portfolio is normalised to ensure that the weights sum to 1 regardless. The optimal w_t is not achievable unless we estimate μ_t and Σ_t .

To control for the out-of-sample estimation errors, we consider the Bayes-Stein shrinkage estimators proposed by Stein (1956) and James and Stein (1961) to build the MV-optimised currency factors. The sample-based MV optimiser relies heavily on historical sample moments and is therefore highly vulnerable to estimation error, particularly in the FX market, where factor returns exhibit time-varying volatility and are subject to structural breaks (Menkhoff et al., 2012a). By shrinking extreme return estimates toward a more stable central mean, the Bayes-Stein framework reduces the overfitting problems inherent in the standard MV optimiser, as demonstrated in the foundational work of Jorion (1986) and further supported by DeMiguel et al. (2009a). This is especially relevant in currency investing, where factors such as carry and momentum are prone to sudden reversals and regime shifts (Lustig et al., 2011).

Empirical evidence further supports the effectiveness of shrinkage in delivering more robust and implementable currency strategies. The theoretical foundations laid out by Black and Litterman (1992) show that blending sample estimates with Bayesian priors leads to more stable and economically sensible portfolio allocations. In the context of currency factors, this translates into reduced portfolio turnover, lower sensitivity to outliers, and improved out-of-sample Sharpe ratios, as shown in the dynamic FX strategy analysis of Della Corte et al. (2009).

Furthermore, the Bayes-Stein estimator enhances the conditioning of the covariance matrix, an important consideration given the instability of factor correlations during periods of market stress (Burnside, 2011). By shrinking the sample covariance matrix toward a structured target (such as a constant-correlation or single-factor model), the estimator avoids the extreme and often impractical weights generated by the standard MV approach, as demonstrated in Ledoit and Wolf (2003). This stabilisation is particularly valuable for the currency factors, where naïve diversification can be undermined by estimation noise (Asness et al., 2013). Overall, the Bayes-Stein approach improves the efficiency and implementability of currency factor strategies, particularly in settings where transaction costs and estimation uncertainty are nontrivial.

Empirically, we follow DeMiguel et al. (2009b) and Platanakis et al. (2021) to implement the Bayes-Stein estimators of Jorion (1986) to optimise our currency factors.⁶ For the estimation parameters, we define R_t as a $N \times 1$ vector of excess returns in month t . Then, the expected returns vector is denoted as μ_t , and Σ_t represents the $N \times N$ estimated out-of-sample covariance matrix, with their sample counterparts given by $\hat{\mu}_t$ and $\hat{\Sigma}_t$, respectively. M refers to the estimation window. The Bayes-Stein estimator of the covariance matrix is estimated as:

$$\Sigma_t = \frac{1}{M - N - 2} \sum_{s=t-M}^{t-1} (R_s - \hat{\mu}_t)(R_s - \hat{\mu}_t)', \quad (8)$$

and the expected return is estimated as:

$$\mu_t = (1 - \hat{\phi}_t) \hat{\mu}_t + \hat{\phi}_t \hat{\mu}_t^{min}, \quad (9)$$

$$\hat{\phi}_t = \frac{N + 2}{(N + 2) + M \times (\hat{\mu}_t - \mu_t^{min})' \Sigma_t^{-1} (\hat{\mu}_t - \mu_t^{min})}, \quad (10)$$

⁶ We are aware of the recent developments in alternative Bayesian approaches such as the hierarchical model of Young and Lenk (1998) and the shrinkage-based estimators of Bodnar et al. (2025). However, for simplicity's sake, we use the most conventional Bayesian approach to demonstrate its effectiveness. Empirical evidence for using alternative estimators can be found in Fan et al. (2025).

where $0 < \hat{\phi}_t < 1$, and $\mu_t^{min} \equiv \hat{\mu}_t' \hat{u}_t^{min}$ is the mean of the sample global minimum variance portfolio excess returns. Following Maurer et al. (2022), we estimate the out-of-sample expected returns and covariance matrix at the end of each month using daily excess returns from the preceding 252 trading days (one year). Portfolio weights are then computed for the following month.

We construct naïve factor portfolios introduced in Section 2.2 by sorting currencies on their respective characteristics and forming high-minus-low portfolios using a 50/50 split. For the MV-optimised factors, we retain the same currencies in the long and short legs as in the naïve counterparts. Within each leg, individual currency weights are constrained to lie between zero and 0.5 to maintain diversification.⁷

The research design is intended to provide a stringent test of the Bayes-Stein framework. Portfolios are formed from a broad cross-section (50% of the currency universe per leg), a setting in which estimation error is known to be particularly severe (DeMiguel et al., 2009b; Papailias et al., 2021; Yuan and Zhou, 2024). In addition, we adopt the conventional in-sample covariance estimator of Jorion (1986) as a conservative approach to control benefits from managing the covariance matrix estimation errors. This setup ensures that any observed outperformance of the MV factors does not arise from limited asset coverage or complex covariance estimation.

Our optimisation framework differs from several existing approaches that aim to enhance the performance of cross-sectional factor portfolios. First, it is distinct from volatility-managed portfolios, which scale portfolio exposure dynamically in response to changes in conditional volatility (e.g., Fleming et al., 2001 and Moreira and Muir, 2017). In these approaches, the underlying signal and portfolio composition remain unchanged, while leverage varies over time to stabilise risk. By contrast, our method does not rely on time-varying leverage. Instead, it directly determines the cross-sectional portfolio weights that maximise mean-variance efficiency across currencies.

Second, our approach differs from enhancing the factor performance by using information from other asset classes. For example, Barroso et al. (2022) studies optimal currency overlay strategies designed to hedge international portfolio exposures, while Kroencke et al. (2014) examines the diversification benefits obtained by combining multiple FX investment styles within a broader portfolio allocation framework. In these settings, optimisation operates at the asset allocation level, determining how investors distribute capital across different asset classes. By contrast, our framework focuses on the construction of the currency factors themselves.

Third, our framework is also distinct from approaches that attempt to improve the predictive signal itself. A large literature enhances factor performance by modifying the signal construction, incorporating additional conditioning variables or redefining characteristics that forecast returns (e.g., Pirrong, 2005 and Fan et al., 2022a). In contrast, our analysis leaves the cross-sectional currency signals unchanged. The improvement arises exclusively from optimally investing in currencies based on their expected returns and covariance structure.

We do not apply our optimisation framework to other widely used currency factors, such as the dollar factor of Lustig et al. (2011) and the Economic/Inflation momentum factor of Dahlquist and Hasseltoft (2020). The dollar factor, as defined by Lustig et al. (2011), is the equally weighted average excess return across all currencies against the US dollar rather than sorting on any currency characteristics. Our optimisation procedure works by reweighting currencies within the long and short legs separately while keeping the total long and short positions equal. Since the dollar factor has no such long-short structure, our procedure does not apply to it. This exclusion is to distinguish our work from three other mean-variance exercises in the literature. The first is to form a mean-variance efficient portfolio across the full set of currencies without using cross-sectional sorting signals, as in Ackermann et al. (2017).⁸ The second is finding the optimal exposure to the dollar factor as a systematic risk factor, which involves treating the dollar factor as a single asset and determining the scalar weight that maximises mean-variance efficiency across it and other style factors. The third is to combine the dollar factor with other currency factors (such as carry, momentum, and value) into a single portfolio and use mean-variance optimisation to find the optimal weight on each factor (Barroso and Santa-Clara, 2015a; Kroencke et al., 2014). Our framework is none of these. Furthermore, the economic and inflation momentum factors are aggregated portfolios constructed from multiple portfolios sorted by various macroeconomic variables, which are not directly compatible with our portfolio optimisation framework. The next section reports the performance of the naïve and optimised factors.

3. Performance comparison: MV vs naïve factors

In this section, we assess the profitability of the MV currency factors and highlight their superiority to the naïve factors and the MV-optimised currency portfolios. As our value factor portfolios are available from the 67th month of the full sample period due to the use of a 5.5-year formation period, we measure the performance of the naïve and MV factor portfolios starting from the 67th month to ensure that all three factors are measured over the same sample period. This results in 429 out-of-sample months, ranging from April 1989 to December 2024.

Table 1 presents an overview of the out-of-sample performance of the naïve and MV factor portfolios, incorporating transaction costs as outlined in Section 2.1. For each factor, we present the performance summaries of both the naïve and mean-variance (MV) portfolios. The “Diff” columns refer to the summary statistics of the return differentials between the naïve and MV factor portfolios. The performance metrics we use are annualised average return (*Mean*), standard deviation (*Vol*), *t*-value, abnormal return (α), *t*-value of the abnormal return ($t(\alpha)$), volatility (*Vol*), Sharpe ratio (*SR*), *t*-value of the Sharpe ratio (*SR.t*), Omega ratio (*Omega*), Dowd ratio

⁷ In practice, we set the lower bound of the weight of each currency to 0.01. If a currency is selected by a factor signal but lacks sufficient return history for mean-variance estimation, we assign it a fixed weight of $\frac{1}{N/2}$, where N is the number of eligible currencies, and optimise the remaining weights over the rest of the portfolio.

⁸ We include this benchmark in our empirical comparison as the “Opt” column in Table 1.

Table 1
Out-of-sample performance using naïve and mean-variance optimised factors.

Strategies	Carry			Momentum			Value			Opt
	naïve	MV	Diff	naïve	MV	Diff	naïve	MV	Diff	
<i>Mean</i>	3.211%	5.656%	2.445%	1.742%	4.431%	2.689%	0.848%	3.535%	2.687%	2.282%
<i>t-value</i>	(3.672)	(4.461)	(2.312)	(1.922)	(3.65)	(2.364)	(1.083)	(2.917)	(2.359)	(2.199)
<i>α</i>	0.030***	0.055***	0.025**	0.018**	0.046***	0.028**	0.007	0.036***	0.028**	0.015**
<i>t(α)</i>	(3.427)	(4.617)	(2.374)	(2.066)	(3.553)	(2.348)	(0.888)	(2.772)	(2.499)	(2.058)
<i>Vol</i>	0.046	0.074	0.028	0.053	0.092	0.039	0.045	0.071	0.026	0.062
<i>SR</i>	0.691***	0.759***	0.068**	0.329**	0.484***	0.155**	0.189	0.499***	0.310**	0.368**
<i>SR.t</i>	(3.941)	(4.361)	(2.148)	(1.974)	(2.885)	(2.259)	(1.131)	(2.966)	(2.258)	(2.122)
<i>Omega</i>	1.693	1.790	0.097	1.285	1.462	0.177	1.150	1.474	0.324	1.346
<i>Dowd</i>	0.125	0.147	0.022	0.065	0.098	0.033	0.035	0.099	0.064	0.062
<i>CER(%)</i>	0.200%	0.400%	0.200%	0.100%	0.200%	0.100%	0.000%	0.200%	0.200%	0.110%
<i>JB-stat</i>	102.2	100.8	154.4	23.8	77.4	219.6	7.9	67.8	136.3	584.2
<i>JB p-value</i>	0.001	0.001	0.001	0.001	0.001	0.001	0.019	0.001	0.001	0.001
<i>Skew</i>	−0.703	−0.414	−0.051	0.211	0.136	−0.413	0.078	0.039	−0.321	−1.057
<i>Kurt</i>	1.935	2.226	2.937	1.075	2.063	3.407	0.647	1.947	2.685	5.312

This table summarises the annualised out-of-sample performance of the naïve and mean-variance factor portfolios with transaction costs from April 1989 to December 2024. Opt represents the mean-variance portfolio across all currencies. Diff refers to the differences between naïve and mean-variance optimised factors' performance. *Mean*, *t-value*, *Vol*, and *Sharpe* denote the annualised average returns, standard deviation, *t*-statistic of mean, and Sharpe ratio of portfolio returns. The *t-value* (in parentheses) uses heteroskedasticity- and autocorrelation-consistent (HAC) standard errors proposed by Newey and West (1987). *SR.t* denotes the *t*-value of the Sharpe ratio based on the bootstrapped Sharpe ratio equality test of Ledoit and Wolf (2008). *Omega*, *Dowd*, and *CER* refer to the Omega ratio, Dowd ratio, and Certainty Equivalent Return, respectively. *JB value* and *JB p-value* represent the value of Jarque-Bera test and the corresponding *p*-value. *Skew* and *Kurt* represent the skewness and excess kurtosis of each return series. *, **, *** represent that the *t*-values are statistically significant at 10%, 5%, and 1% levels.

(Dowd) and Certainty Equivalent Rate (CER).⁹ We further apply the Jarque-Bera test of Jarque and Bera (1980) to test the normality of each factor return series, where the results are presented as the Jarque-Bera value (*JB-value*), the *p*-statistic (*JB p-value*), as well as the skewness (*Skew*) and excess kurtosis (*Kurt*). The optimal portfolio (Opt) in the last column is reported only as an unconditional mean-variance benchmark across all currencies.¹⁰

For the naïve portfolio with equally weighted schemes in both the long and short legs, the carry factor performs best among the three factors, with an average annual return of 3.211%, an abnormal return of 0.030, and a Sharpe ratio of 0.691. Following closely, the momentum factor displays the second-highest profitability, with an average return of 1.742%, an α of 0.018, and a Sharpe ratio of 0.329. Notably, the carry and momentum factors report statistically significant profits at the 1% and 10% levels, respectively. In contrast, the naïve value factor generates an insignificant annual return of 0.848% and an insignificant α of 0.007. Furthermore, the carry factor reports the best OOS performance with the highest Omega ratio (1.693), Dowd ratio (0.125), and CER (0.223%).

Remarkably, the out-of-sample performance improves considerably upon incorporating the MV factors. Table 1 reveals that the MV factors consistently yield a mean return that is at least 2.445% (in the case of carry) higher than their naïve counterparts. The most improved factor, momentum, stands out with impressive growth in annual return, increasing to 4.431% after applying the MV optimisation, versus 1.742% following the naïve approach. Furthermore, the optimised value portfolio even leads to an increase of 2.687% per annum. The superiority of the MV factors in average returns and abnormal profits is statistically significant at the 5% level across all three factors.

Although the outperformance of the MV factors is often accompanied by higher return volatility, the MV factors still exhibit higher risk-adjusted performance in comparison with the naïve one. The increases in the Sharpe ratios are at least 0.068 across the three factors. We measure the *t*-values of the Sharpe ratios based on the bootstrapped Sharpe ratio equality test of Ledoit and Wolf (2008) to mitigate the impacts of the outliers. Such a method can ensure that the superiority in SR is consistent across the entire sample period, rather than relying on certain sub-periods. The *t*-values of the Sharpe ratios indicate that the risk-adjusted returns for all three factors are significantly improved at the 5% level. Notably, the MV value factor demonstrates the most pronounced improvement in the Sharpe ratio, rising from 0.189 to 0.499.

Improvements are observed for the Omega ratios across all three factors, with the most significant change observed in the value factors (increasing from 0.150 to 1.474). Similarly, the Dowd ratios for the MV factors exceed those of their naïve counterparts. Meanwhile, the positive difference in CER further echoes the MV factors' outperformance compared to their naïve counterparts.

Moreover, the distributional statistics indicate that both the naïve and MV-optimised factor returns deviate from normality. The Jarque-Bera (JB) test strongly rejects the null hypothesis of normality for all factor portfolios at the 5% level. This result is consistent with the skewness and excess kurtosis reported in the lower panel of the table. In particular, the carry factors exhibit negative skewness and substantial excess kurtosis, indicating asymmetric return distributions with fat tails. These distributional characteristics are consistent with the well-documented crash risk in currency strategies, whereby high-yielding currencies occasionally experience sharp

⁹ Formal definitions of all performance metrics appear in Appendix B.

¹⁰ In line with our MV factors, we construct the mean-variance optimised portfolio using Bayes-Stein estimators for both expected returns and covariance matrices with a rolling one-year formation period. We impose the same weight constraints such that individual currency allocations are bounded between zero and 0.5.

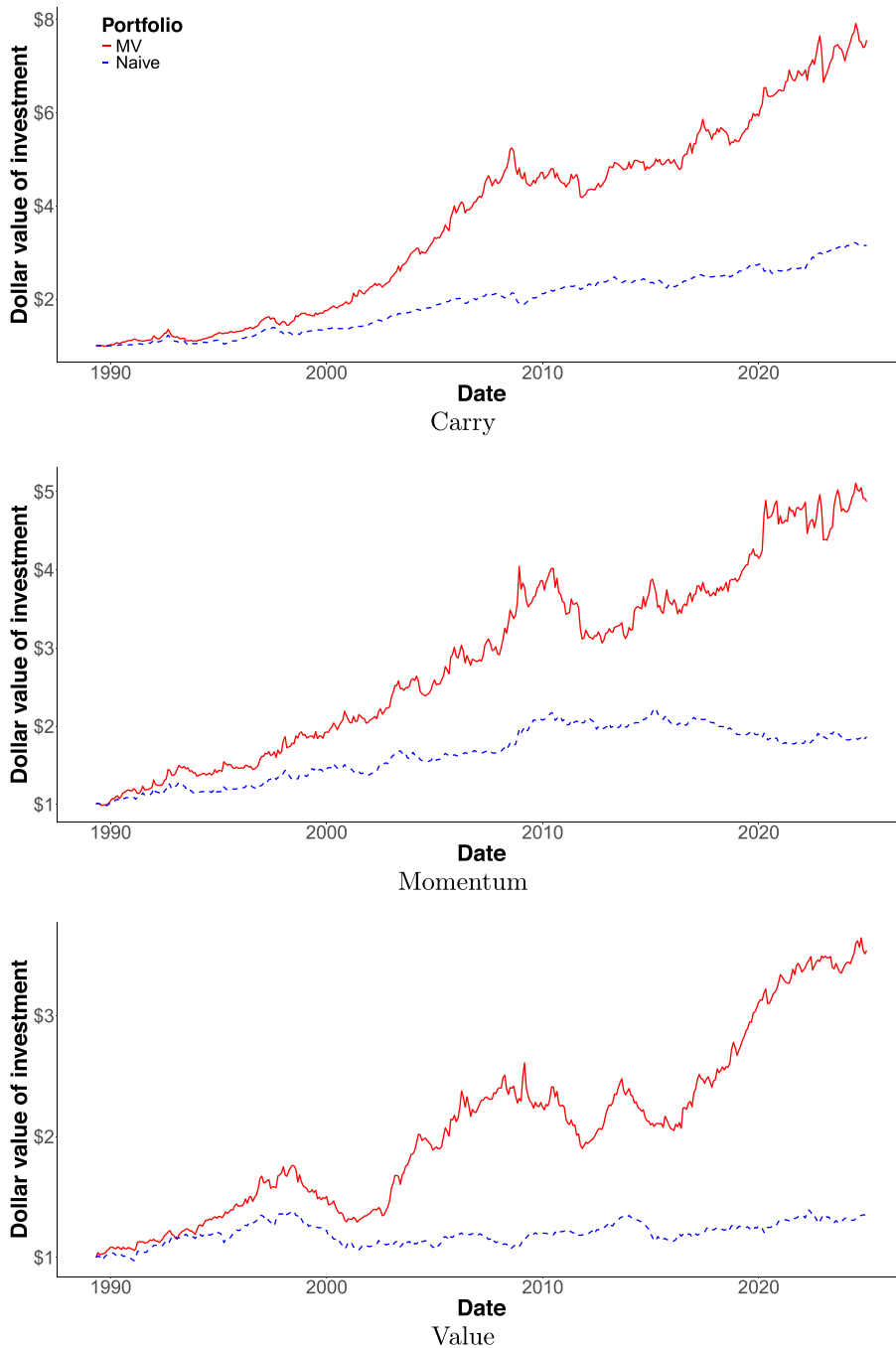


Fig. 1. Cumulative performance of the currency factor portfolios.

These plots exhibit the cumulative performance of the MV-optimised and the naïve factor portfolios. We assume \$1 is invested in each portfolio at the beginning of the sample period. MV and naïve refer to the mean-variance diversified factors and the original equally weighted factors, respectively. The sample ranges from April 1989 to December 2024.

reversals during periods of financial stress (Brunnermeier et al., 2008). Compared with the naïve portfolios, the MV-optimised factors generally display skewness closer to zero, suggesting that the optimisation framework produces return distributions that are more symmetric. Although mean-variance optimisation is derived under normality, rejection of this assumption does not invalidate the approach. Hanoch and Levy (1969) show that mean-variance analysis remains a valid approximation to expected utility maximisation for a wide class of utility functions under moderate departures from normality. The superior out-of-sample performance of the MV-optimised factors documented above is consistent with this result and accords with the finding that mean-variance rules are difficult to outperform even when returns exhibit non-normal features (DeMiguel et al., 2009b; Platanakis et al., 2021).

Fig. 1 plots the cumulative performance of the MV-optimised and naïve factor portfolios over the investment horizon, after accounting for transaction costs. The figure highlights that the MV-optimised portfolios (red lines) deliver substantially superior cumulative returns compared to the original naïve factor portfolios (blue dashed lines), consistent with the results reported in Table 1.

For all three factors, the cumulative investment values of the MV-optimised and naïve portfolios track each other closely before 2000, and the MV portfolios exhibit modest but consistent outperformance during this period. From that point onwards, the MV portfolios exhibit a marked upward trajectory, far outpacing their naïve counterparts. In the case of the carry factor, the MV portfolio demonstrates especially strong growth after 2003, with relatively smooth upward movement interspersed with minor drawdowns, ultimately reaching over \$7.5 by the end of 2024, while the naïve carry portfolio remains below \$2.5. Similarly, for the momentum factor, the MV portfolio displays pronounced and persistent outperformance since the beginning of the investment horizon. The naïve momentum strategy, though upward trending, remains much flatter and ends with a significantly lower terminal value than the MV portfolio construction. Furthermore, the MV-optimised value portfolio also begins to outperform the naïve version starting in 2000, indicating diversification benefits under the MV scheme. Despite the considerable volatility during the global financial crisis of 2007–2008, the MV value portfolio maintains its positive cumulative return, ultimately closing above \$3.5. Conversely, the naïve value portfolio experiences repeated drawdowns and stagnation, achieving a marginal positive cumulative return throughout the sample period.

In terms of risk management, the MV-optimised portfolios often exhibit higher volatility compared to the naïve diversification of factor portfolios, especially in the cases of carry and value. Nevertheless, these portfolios demonstrate faster and more sustained recoveries relative to their naïve counterparts, highlighting the resilience and adaptive capacity of the MV optimisation framework under market panic conditions. To ensure that the outperformance of the MV-optimised factors is not driven by thinly traded currencies, we provide a robustness analysis using a smaller and more liquid sample of 19 currencies as in Ackermann et al. (2017). The results of the robustness analysis remain consistent with those of the main sample. We report them in Section 6.3.

The correlation structure in Table 2 first shows that the MV-optimised factors remain meaningfully related to their naïve counterparts, but are clearly not identical to them. Panel A shows that each optimised factor remains positively correlated with its naïve counterpart, indicating that the optimisation procedure preserves the underlying economic signals embedded in the traditional currency factors. This pattern also appears in all three subsamples. For example, the carry/optimised-carry correlation ranges from 0.331 to 0.569, the momentum pair from 0.518 to 0.791, and the value pair from 0.250 to 0.452.

Another notable result is that optimisation substantially increases comovement across the style factors. In the full sample, the correlation between naïve carry and naïve momentum is only 0.024, whereas the correlation between optimised carry and optimised momentum rises sharply to 0.679. Likewise, the naïve carry–value correlation is only 0.033, while the optimised carry–value correlation is 0.394; the naïve momentum–value correlation is -0.263 , but the corresponding optimised momentum–value correlation becomes 0.400. These results suggest that the optimisation procedure extracts common priced components of currency risk that are not captured by the characteristic-sorted portfolios.

Meanwhile, the correlation matrix reveals important diversification properties. Several optimised factors exhibit relatively low or even negative correlations with the dollar factor. For instance, the optimised momentum factor displays a modest negative correlation with the dollar factor in the full sample and in multiple subsamples. Such patterns suggest that the optimised factors may provide useful hedging opportunities for investors exposed to broad dollar movements. In particular, strategies that combine optimised currency factors with dollar exposure may achieve improved portfolio diversification relative to combinations based solely on naïve factors.

4. Dissecting the outperformance

4.1. A decomposition model

To further understand the source of the outperformance of the MV-optimised factors, we adopt the expected return model of Lo and MacKinlay (1990) and Lewellen (2002), where the expected return function of a univariate-sorted portfolio across N currencies is:

$$E[\pi_t^p] = \frac{1}{N} \sum_{i=1}^N \text{cov}(r_{i,t}^{\text{signal}}, r_{i,t}) - \text{cov}(r_{-t}^{\text{signal}}, \bar{r}_t) + \frac{1}{N} \sum_{i=1}^N (\mu^i - \bar{\mu})^2, \quad (11)$$

where $r_{i,t}$ represents the excess return of currency i in month t , $r_{i,-t}^{\text{signal}}$ denotes the signal return of currency i over the formation period, $\bar{r}_{-t}^{\text{signal}}$ is the cross-sectional average of all currencies' signal returns over the formation period, and μ^i represents the unconditional expected return of currency i .¹¹ This model demonstrates that the expected returns of a currency factor portfolio can be decomposed into three components:

- 1) $\frac{1}{N} \sum_{i=1}^N \text{cov}(r_{i,-t}^{\text{signal}}, r_{i,t})$ reflects the correlation between currency excess returns and the signal returns over the formation period, i.e., the time series predictability.

¹¹ We simplify Eq. (11) by adopting the assumption of Maurer et al. (2022) that the expected signal return is equal to the expected return of currency. The last term should be as $\frac{1}{N} \sum_{i=1}^N (\mu^{\text{signal}} * \mu^i) - \mu^{\text{signal}} * \bar{\mu}$, where μ^{signal} denotes the unconditional expected signal return of currency i . We adopt this accurate measurement to conduct the empirical results. To ensure the results are comparable, we also use the negative value signal returns in the decomposition to hold a consistent time-series prediction across three factors.

Table 2
Dynamic correlation across currency factors.

Factor	Carry	Opt.carry	Momentum	Opt.mom	Value	Opt.value	Dollar
Panel A: entire sample							
Carry	–	0.453	0.024	0.005	0.033	0.033	0.263
Opt.carry	0.453	–	0.329	0.679	–0.241	0.394	0.096
Momentum	0.024	0.329	–	0.650	–0.263	0.216	–0.038
Opt.mom	0.005	0.679	0.650	–	–0.321	0.4	–0.091
Value	0.033	–0.241	–0.263	–0.321	–	0.332	0.164
Opt.value	0.033	0.394	0.216	0.400	0.332	–	–0.023
Dollar	0.263	0.096	–0.038	–0.091	0.164	–0.023	–
Panel B: April 1989 to December 1999							
Carry	–	0.569	0.164	0.079	–0.007	0.029	0.212
Opt.carry	0.569	–	0.444	0.651	–0.232	0.269	0.344
Momentum	0.164	0.444	–	0.791	–0.262	0.24	0.133
Opt.mom	0.079	0.651	0.791	–	–0.325	0.336	0.174
Value	–0.007	–0.232	–0.262	–0.325	–	0.452	–0.069
Opt.value	0.029	0.269	0.240	0.336	0.452	–	0.054
Dollar	0.212	0.344	0.133	0.174	–0.069	0.054	–
Panel C: January 2000 to December 2010							
Carry	–	0.464	–0.049	0.035	–0.017	0.039	0.337
Opt.carry	0.464	–	0.252	0.700	–0.274	0.479	–0.061
Momentum	–0.049	0.252	–	0.640	–0.215	0.174	–0.127
Opt.mom	0.035	0.700	0.640	–	–0.267	0.466	–0.272
Value	–0.017	–0.274	–0.215	–0.267	–	0.25	–0.028
Opt.value	0.039	0.479	0.174	0.466	0.250	–	–0.247
Dollar	0.337	–0.061	–0.127	–0.272	–0.028	–0.247	–
Panel D: January 2011 to December 2024							
Carry	–	0.331	–0.144	–0.107	0.133	0.034	0.252
Opt.carry	0.331	–	0.280	0.683	–0.213	0.394	0.045
Momentum	–0.144	0.280	–	0.518	–0.33	0.264	–0.17
Opt.mom	–0.107	0.683	0.518	–	–0.374	0.383	–0.136
Value	0.133	–0.213	–0.330	–0.374	–	0.334	0.577
Opt.value	0.034	0.394	0.264	0.383	0.334	–	0.204
Dollar	0.252	0.045	–0.170	–0.136	0.577	0.204	–

This table reports pairwise correlations among the naïve and mean–variance (MV) optimised currency factors and the dollar factor. Carry, Momentum, and Value denote the standard characteristic-sorted currency factors constructed using equal-weighted portfolios. Opt.carry, Opt.mom, and Opt.value represent the MV-optimised factors. Dollar denotes the dollar factor capturing the average excess return across currencies following [Lustig et al. \(2011\)](#). Panel A reports correlations for the full sample, while Panels B–D present results for three subperiods: April 1989 to December 1999, January 2000 to December 2010, and January 2011 to December 2024. All correlations are computed using monthly excess returns.

- 2) $-cov(r_{-t}^{signal}, \bar{r}_t)$ captures the diversification-benefit channel, where a past high currency signal is associated with low returns on other currencies.
- 3) $\frac{1}{N} \sum_{i=1}^N (\mu^i - \bar{\mu})$ captures the signal-amplification channel generated by dispersion in persistent mean returns.

This decomposition reflects that the expected returns of a cross-sectional factor portfolio are sourced from both the time series predictability of the past signals, i.e., Component 1), and the cross-sectional predictability, i.e., Component 2) and 3).

We start by investigating the time series component and hypothesise that the MV weights jointly improve the time series predictability of currency characteristics. To test this hypothesis, we further apply pooled panel regressions:

$$r_{i,t} = \alpha + \beta_1 r_{i,t-1}^{signal} + \beta_2 w_{i,t-1}^{signal} + \beta_3 r_{i,t-1}^{signal} * w_{i,t-1}^{signal} + \epsilon_t, \quad (12)$$

where $w_{i,t-1}^{signal}$ denotes the weight of currency i determined at the end of month $t - 1$. Other notations are as described in Eq. (11).

Table 3 presents the regression results of Eq. (12) to examine the time series predictability of currency weights for three factors. Interestingly, the optimised currency weights generally do not enhance the time series predictability of the currency characteristics. For all three factors, columns 3, 6, and 9 show that the interaction terms, $Carry * W^{carry}$, $Mom * W^{mom}$, and $Value * W^{value}$ yield statistically insignificant coefficients. This suggests that the interactions between the factors and their optimised weights do not provide any additional predictive power for future currency returns. The insignificance of these interaction terms also implies that the optimised weights cannot effectively capture time-varying predictability in currency returns, even though the currency carry,

Table 3
Time series predictability of the currency weights.

Model	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
<i>Carry</i>	0.830*** (2.60)		0.816*** (2.40)						
<i>W^{carry}</i>		0.005*** (3.18)	0.003* (1.78)						
<i>Carry * W^{carry}</i>			−0.156 (−0.25)						
<i>Mom</i>				0.350*** (21.81)		0.358*** (22.00)			
<i>W^{mom}</i>					0.005*** (3.15)	−0.012*** (−8.45)			
<i>Mom * W^{mom}</i>						0.035 (0.74)			
<i>Value</i>							0.010*** (7.35)		0.015*** (8.40)
<i>W^{value}</i>								0.004*** (3.28)	0.009*** (5.90)
<i>Value * W^{value}</i>									−0.003 (−0.42)
Currency fixed effect	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Month fixed effect	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Number of currencies	48	48	48	48	48	48	48	48	48
Adj. <i>R</i> ²	0.379	0.373	0.379	0.621	0.374	0.624	0.396	0.398	0.404

This table reports the results of panel regressions testing the joint predictability between currency characteristics and the mean-variance portfolio weights. To ensure the results are comparable, we use the negative value signal returns in the regression model. *W^{carry}*, *W^{mom}*, and *W^{value}* represent the weights of the optimised carry, momentum, and value portfolios. Models (1) to (3) show the predictability of the currency carry. Models (4) to (6) show the predictability based on the currency's past returns. Models (7) to (9) show the predictability based on the currency value. The *t*-value (in parentheses) is calculated based on the robust standard error of Thompson (2011). The number of observations and the adjusted *R*² are also reported. *, **, *** represent that the *t*-values are statistically significant at 10%, 5% and 1% levels, respectively.

momentum, and value signals themselves report positive and highly significant coefficients of 0.816 (*t* = 2.40), 0.358 (*t* = 22.00), and 0.015 (*t* = 8.40), respectively.

In the second step of our analysis, we examine which cross-sectional components primarily drive the performance gains from optimisation. The decomposition of strategy returns reveals substantial heterogeneity in the contribution of each component. Table 4 reports the performance of the three components for the carry, momentum, and value strategies under both naïve and optimised portfolio constructions. Panel A presents the annual mean of the excess returns. Panel B shows the abnormal return (α) estimated by:

$$R_t^c = \alpha + R_t^{Dollar} + \epsilon, \quad (13)$$

where R_t^c denotes the returns of a given component, and R_t^{Dollar} refers to the dollar factor of Lustig et al. (2011).

In Panel A, for component (1), which captures the weighted average of individual expected excess returns, the differences between naïve and optimised portfolios are generally statistically insignificant for the carry factor and statistically negative at the 1% level for momentum and value. These findings are consistent with the panel regression results in Table 3, which indicate that the MV-optimised weighting scheme does not improve the time-series predictability of currency characteristics.

The notable gains from optimisation arise in component (2), which we interpret as the diversification benefit: by tilting weights toward currencies whose returns co-move less strongly with the rest of the portfolio, the optimiser converts the cross-sectional covariance structure into higher portfolio returns. This component exhibits the most considerable economic differences between the naïve and MV-optimised portfolios across three components. For the carry factor, the MV-optimised portfolio increases the return contribution from 0.60% to 2.84%, a difference of 2.20%. The momentum factor shows a dramatic improvement, with the component (2) return rising from −11.85% in the naïve case to 1.04% under the MV case, producing a statistically significant difference of 12.66% (*t*-statistic = 5.60). Similarly, the value strategy sees a significant enhancement of 5.85% (*t*-statistic = 3.17).

Component (3) captures a conceptually distinct signal-amplification channel: the optimiser tilts capital toward currencies with persistently higher unconditional expected returns and away from those with lower expected returns, thereby amplifying the cross-sectional return signal. For all factors, the improvements in this component are statistically significant at the 1% level, suggesting that MV optimisation more effectively exploits persistent return differentials across currencies. Although both components contribute

Table 4
Performance of decomposed components.

Component	Naïve			Optimised			Difference		
	Carry	MOM	Value	Carry	MOM	Value	Carry	MOM	Value
Panel A: excess return									
1)	2.423% (3.24)	13.584% (17.11)	3.342% (4.21)	1.667% (0.97)	1.995% (1.35)	−1.401% (−0.99)	−0.808% (−0.50)	−12.006% (−6.73)	−4.985% (−3.07)
2)	0.597% (1.00)	−11.853% (−14.01)	−2.133% (−1.70)	2.845% (1.60)	1.042% (0.45)	4.067% (2.35)	2.197% (1.24)	12.664% (5.60)	5.851% (3.17)
3)	0.026% (7.43)	−0.466% (−7.35)	−0.776% (−3.22)	0.052% (13.03)	0.031% (4.00)	−0.031% (−1.02)	0.026% (5.09)	0.496% (8.16)	0.724% (3.04)
Panel B: abnormal return (α)									
1)	0.170% (2.99)	1.139% (18.91)	0.271% (4.04)	0.072% (0.72)	0.101% (0.95)	−0.053% (−0.48)	−0.101% (−0.88)	−1.075% (−8.51)	−0.345% (−2.89)
2)	0.072% (1.48)	−0.994% (−15.61)	−0.201% (−1.74)	0.322% (3.35)	0.193% (1.23)	0.280% (1.99)	0.245% (2.42)	1.175% (7.45)	0.450% (2.80)
3)	0.002% (6.00)	−0.041% (−6.23)	−0.059% (−2.87)	0.004% (6.18)	0.003% (3.10)	−0.003% (−1.03)	0.002% (2.58)	0.043% (7.06)	0.055% (2.78)

Panels A and B in this table report the factor log returns per annum and the abnormal return (α) created by each component, respectively. The last three columns present the differences in log returns between the naïve and optimised components. Components 1), 2), and 3) represent the correlation between currency excess returns and the signal returns, the diversification-benefit channel generated by cross-covariances, and the signal-amplification channel generated by the dispersion of unconditional expected returns, respectively. The *t*-value (in parentheses) uses heteroskedasticity- and autocorrelation-consistent (HAC) standard errors proposed by Newey and West (1987). *, **, *** represent that the *t*-statistics are statistically significant at 10%, 5% and 1% levels, respectively.

positively to the cross-sectional gains, their economic mechanisms differ: component (2) operates through the covariance structure, determining which currencies to combine, while component (3) operates through the dispersion of unconditional means—determining which currencies earn higher long-run premia. Currency markets, with their small cross-section and comparatively well-estimated covariance matrices, are well suited to exploiting both channels.

Panel B reports the abnormal returns (α) for each component and supports the evidence in Panel A. Component (1), which captures the weighted average of individual expected excess returns, shows little benefit from optimisation. The carry difference is small and insignificant, while the momentum and value differences are significantly negative. This pattern indicates that the MV weights do not improve the time-series predictability of the underlying signals. By contrast, component (2), the diversification-benefit channel, explains most of the performance improvements by exploiting lower comovement among currencies. Across three factors, the differences are at least 0.25%, and all are statistically significant at the 5% level. Component (3), the signal-amplification channel linked to the dispersion of unconditional mean returns, also contributes positively, with significant increases at the 1% level.

In summary, the decomposition results indicate that the outperformance of the MV-optimised factors primarily arises from cross-sectional portfolio construction rather than improvements in time-series predictability. Such gains are mainly driven by two economically distinct channels: the diversification-benefit and signal-amplification. The optimal weighting scheme exploits the cross-sectional properties of currency excess returns by assigning lower weights to currencies that co-move closely with other positions and higher weights to those that offer stronger diversification benefits.

From an economic perspective, the decomposition reveals two sources of outperformance for the MV factors. The first source is covariance management. By tilting portfolio weights toward currencies that offer stronger diversification benefits, the MV portfolios convert unfavourable comovement patterns into higher returns through component (2). The second source is the dispersion in expected returns across currencies. The MV weighting scheme systematically allocates more capital to currencies with persistently higher average returns and reduces exposure to those with lower expected returns, thereby increasing the contribution from component (3). By contrast, component (1), which reflects the weighted average of individual expected excess returns, contributes little to the overall improvement and is occasionally negative. This pattern suggests that optimisation does not enhance the time-series predictability of the currency characteristics; instead, the economic gains arise primarily from more efficient cross-sectional portfolio construction. This finding is consistent with the notion that currency investors price systematic rather than transitory fluctuations, so that portfolio improvements arise from better capturing cross-sectional risk premia rather than short-term return dynamics (Lustig et al., 2011; Atanasov and Nitschka, 2015).

However, the effectiveness of this allocation mechanism may weaken when return differentials across currencies are less pronounced, such as in periods when currency returns become highly correlated. We examine this possibility in the next subsection.

4.2. Does international diversification matter?

Our decomposition analysis reveals that the outperformance of the optimised factors is driven primarily by enhanced cross-sectional predictability rather than by improved time-series forecasting. The optimal weighting schemes effectively exploit both signal amplification from the dispersion of persistent mean returns and diversification benefits from cross-covariances, an advantage that diminishes when currency returns are highly correlated. This prompts an economic investigation into the source of this exploitable

heterogeneity in returns. We posit that international diversification is the key channel, grounded in the structural features of the global currency market.

A distinguishing feature of the currency market is the natural segmentation between developed and emerging currencies, which efficiently reflects variations in macroeconomic fundamentals. This segmentation has important implications for international diversification. Currencies from emerging regions are typically associated with higher macroeconomic volatility, less efficient pricing, and lower correlation with developed markets (Lustig et al., 2011; Menkhoff et al., 2012a), which may enhance cross-sectional predictability and diversification benefits. Conversely, developed currencies exhibit more integrated markets and lower return dispersion, potentially limiting the gains from optimisation. These features align with asset pricing theories where global diversification benefits are most pronounced when market segmentation or structural heterogeneity produces low return comovement across regions (Bekaert and Harvey, 1995; Karolyi and Stulz, 1996). Therefore, we hypothesise that the superiority of MV optimisation is related to international diversification caused by the fundamental variation of emerging and developed currencies.

To validate this hypothesis, we divide our sample into two sub-samples: currencies from developed and emerging economies, based on the definition provided by the International Monetary Fund (IMF). 29 out of 48 are considered as developed currencies, including those of Australian Dollar, Austrian Schilling, Belgian Franc, Canadian Dollar, Cypriot Pound, Czech Koruna, Danish Krone, Finnish Markka, French Franc, German Mark, Greek Drachma, Hong Kong Dollar, Icelandic Krona, Irish Pound, Israeli Shekel, Italian Lira, Japanese Yen, Dutch Guilder, New Zealand Dollar, Norwegian Krone, Portuguese Escudo, Singapore Dollar, South Korean Won, Spanish Peseta, Swedish Krona, Swiss Franc, New Taiwan Dollar, and UK Pound, with the remainder designated as emerging currencies.¹²

In the previous section, the cumulative performance in Fig. 1 highlights a regime change in the late 90s/early 2000s when the outperformance of the optimised carry and momentum strategies became larger thereafter. This change corresponds to the launch of the Euro in 1999, when ten European currencies were replaced. Meanwhile, in the early 2000s, there was a sharp increase in the number of emerging currencies. This creates a more favourable investment environment for international diversification purposes. Fig. 2 plots the number of available currencies (solid red line) and developed currencies (dashed blue line) during our sample period. Prior to the launch of the Euro, the sample consisted mainly of developed currencies (16 out of 18 currencies), suggesting insufficient international diversification for portfolio construction.

Next, to formally assess the effect of international diversification, we analyse the time-varying average correlation across different groups of currencies. Following Pollet and Wilson (2010), the sample correlation for currencies j and k , $\rho_{jk,t}$, is measured as:

$$\rho_{jk,t} = \frac{\sigma_{jk,t}}{\sigma_{j,t}\sigma_{k,t}}, \quad (14)$$

where $\sigma_{jk,t}$ represents the covariance for currencies j and k , $\sigma_{j,t}$ and $\sigma_{k,t}$ denotes the standard deviation of currency j and k respectively. We adopt the daily excess returns in month t to estimate $\sigma_{jk,t}$, $\sigma_{j,t}$ and $\sigma_{k,t}$. Then, the average correlation is measured as follows:

$$AC_t = \sum_{j=1}^N \sum_{k \neq j} w_{j,t} w_{k,t} \rho_{jk,t}, \quad (15)$$

where N currencies are available.

The time-varying average correlation shown in Fig. 3 examines the consistency of currencies' comovement for both the developed and emerging samples. Throughout the sample period, emerging currencies exhibit a much lower average correlation against the US Dollar than developed currencies. The average AC for emerging currencies for the entire sample period is 0.108, significantly lower than the AC of 0.212 for developed currencies. This gap is only slightly narrowed around the 2007–2008 financial crisis period. Therefore, we argue that the superior performance of the MV optimiser in estimating currency returns is attributable to the additional covariance information derived from international diversification.

4.3. Distribution of currency α and optimisation

The low correlation between currencies caused by international diversification leads to variations in their return distributions, which can materially affect the performance of MV-optimised portfolios (Hanoch and Levy, 1969). To further explore the underlying mechanism behind the superiority of the MV factor portfolios, we link their outperformance to the distribution of abnormal returns and idiosyncratic volatility (IVOL) of currencies, using a framework proposed by DeMiguel et al. (2009b) and Platanakis et al. (2021). This approach provides an attribution analysis of the outperformance by examining abnormal returns and IVOL calculated from established asset pricing models. Specifically, we adopt a currency market version of the factor model proposed by DeMiguel et al. (2009b) and Platanakis et al. (2021) and estimate the abnormal returns and IVOL for each currency as:

$$rx_{k,t} = \alpha_k + \sum \beta \times r_{factor,t} + \epsilon_{t,k}, \quad (16)$$

where $rx_{k,t}$ is the excess return of currency k in month t ; α_k represents the abnormal return of currency k ; β refers to the factor loading; r_{factor} refers to the excess returns of the control factors, including the US dollar index (DXY), carry, momentum, and value factors; and $\epsilon_{t,k}$ is the noise term and is assumed to be distributed as $\epsilon_{t,k} \sim N(0, \sigma_{\epsilon,k})$ with $\sigma_{\epsilon,k}$ representing the idiosyncratic volatility (IVOL).

¹² The IMF designates Croatia, Slovakia, and Slovenia as advanced economies in 2023, 2009, and 2007, respectively. Their currencies are excluded from the developed sub-sample before they attain advanced economic status.

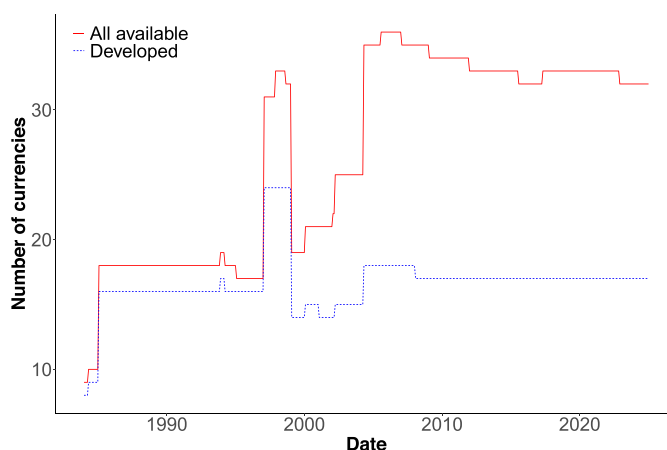


Fig. 2. Number of available currencies.

This plot exhibits the number of currencies available to investors. The blue dashed line represents the number of currencies from the developed economies across the sample period, with the solid red line represents the total number of currencies available.

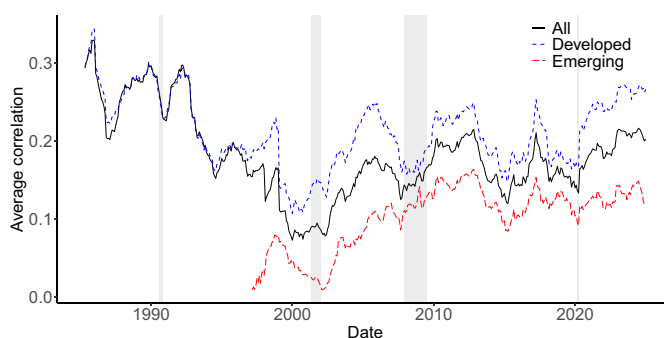


Fig. 3. Average pairwise correlation.

This plot displays the average pairwise correlation for the full sample and the developed and emerging sub-samples over time. The sample period is April 1989 to December 2024. The average correlation is calculated as the equally weighted average of pairwise correlations of monthly currency excess returns over the 60-month formation period. For the emerging sample, we omit the period before 1997 as it consists of only two currencies. The shaded ranges represent the recession periods identified by the NBER.

Fig. 4 plots the distribution of α_k for both the developed and emerging currencies over the sample period. The means of α_k in both the developed (dashed blue line) and emerging (dashed red line) samples are negative but very close to zero, with the mean in the emerging sample being slightly lower than in the developed sample. Due to the visible fat tails, the standard deviation of α_k in the emerging sample is higher than in the developed sample. Previous literature, e.g., DeMiguel et al. (2009b), showed that the optimal portfolios fail to significantly outperform the 1/N rule based on a simulated return series with the zero-alpha assumption, which assumes all asset alphas are precisely zero. However, as argued by Jarrow (2010), while the zero-alpha assumption may be useful for academic study, it is too rigorous empirically. Platanakis et al. (2021) reveal that the performance of optimal portfolios is subject to the standard deviation of α_k across the available assets over the estimation window. Therefore, they fit α_k to a normal distribution with a mean of zero and a standard deviation of 30 basis points (i.e., the zero-mean assumption), finding that the optimal portfolios significantly outperform the naïve ones. We empirically test the abnormal returns for each currency and find that most of their abnormal returns are insignificantly different from zero. However, average standard deviations are at 12.4% for developed currencies and 29.6% for emerging currencies. The results of an F -test indicate that the variance of α_k in both samples is significantly different from zero at the 1% level. These results are reported in Appendix C.

There have been debates on whether the relationship between the outperformance of optimisation and the standard deviation of abnormal returns is conditional on the level of idiosyncratic volatility over the estimation window. Under the zero-alpha assumption, DeMiguel et al. (2009b) show that the outperformance of the optimised portfolio increases with the average IVOL across assets. In contrast, based on simulations, Platanakis et al. (2021) find that for a given level of abnormal return dispersion, higher IVOL reduces the benefits of optimisation. Jointly, the positive effect of abnormal return dispersion on portfolio performance diminishes as average IVOL rises.

Our empirical finding based on currency data is consistent with the zero-mean assumption of Platanakis et al. (2021). Specifically, we examine whether the standard deviation of α_k and the average IVOL jointly influence the performance of MV factor portfolios. We developed two hypotheses as follows:

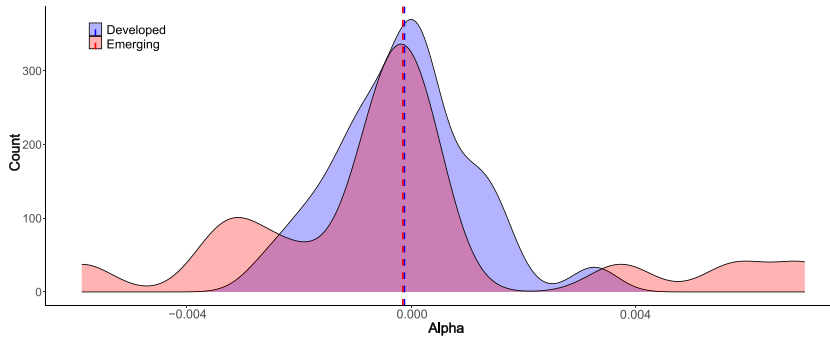


Fig. 4. Distribution of abnormal returns (α_k) for developed and emerging currencies.

This figure shows the density-fitted areas of distributions of abnormal returns. The dashed lines indicate the mean of each distribution. The blue area represents the distribution of abnormal returns for the developed sub-sample, while the red area denotes the distribution based on the emerging sub-sample.

- *H1*: The enhanced performance of the MV factor portfolios is not derived from an increase in the standard deviation of currencies' abnormal profits.
- *H2*: The enhanced performance of the MV factor portfolios is not derived from low average IVOL.

To test our hypotheses, we first estimate the time-varying standard deviation of α_k and average IVOL, $\sigma_{\epsilon,k}$, over the entire sample period. We implement a 252-day rolling window, in line with the estimation windows of our Bayes-Stein estimators, to estimate α_k and $\sigma_{\epsilon,k}$ over the estimation window for each month. Panel A of Fig. 5 plots the time-dynamic standard deviations of α_k for the full (solid green line), developed (dotted blue line), and emerging (dashed red line) samples. Before 1999, the red line is omitted due to an insufficient number of emerging currencies. Afterwards, the standard deviation of α_k for the emerging sample spiked around 2001 and 2008, with the red line moving far above the blue line over most periods thereafter. Overall, the standard deviation of α_k in the emerging sample is much higher than that of the developed sample. Panel B of Fig. 5 presents the time series evolution of average IVOL for the full, developed, and emerging currency samples. Throughout most of the sample period, the average IVOL for the emerging market group (red line) remains higher compared to developed and full samples, displaying greater currency-specific risks. Notably, all three series exhibit pronounced spikes during the 2001 dot-com bubble and the 2007–2008 global financial crisis, consistent with increased idiosyncratic risk during periods of crises.

Next, we regress the outperformance of the optimised factor portfolios on the lagged time-varying standard deviation of abnormal returns and the average IVOL, and control for the joint effect, given as:

$$r_{t+1}^{diff} = \alpha_{diff} + \beta_1 SD_{t,p} + \beta_2 AIV_{t,p} + \beta_3 (SD_{t,p} \times AIV_{t,p}) + \sum \beta \times r_{t,control} + \epsilon_p, \quad (17)$$

where r_{t+1}^{diff} denotes the return difference in month $t + 1$ between the MV-optimised factor and the naïve factor. The key explanatory variables are $SD_{t,p}$, the cross-sectional standard deviation of abnormal currency returns, and $AIV_{t,p}$, the average IVOL, measured as the mean of $\sigma_{\epsilon,k}$ across currencies within sample p . Both $SD_{t,p}$ and $AIV_{t,p}$ are constructed using the residuals and abnormal returns estimated from Eq. (16), based on one of the three currency samples, full, developed: or emerging markets, denoted by subscript p . The interaction term, $SD_{t,p} \times AIV_{t,p}$, captures the joint effect of dispersion in abnormal returns and the average IVOL on the MV portfolio's outperformance. All independent variables are lagged by one month relative to r_{t+1}^{diff} , since the MV portfolios use only information available until time t to determine asset weights in month $t + 1$. To control for the market and macroeconomic dynamics, we include the economic- and inflation-momentum factors of Dahlquist and Hasseltoft (2020) as the additional control variables.

Table 5 presents the regression estimates of Eq. (17), reporting the coefficients (β) and corresponding t -statistics for each MV-optimised factor portfolio. The regressions examine whether the outperformance of the MV factors in month $t + 1$ is linked to the time- t standard deviation of abnormal returns, α_k , the average IVOL, as well as the interaction between these two variables.

In the full sample, the coefficients on β_1 are positive for all three factors, with the carry and momentum factors being statistically significant at the 5% and 10% levels, respectively. These results imply that the MV carry and momentum portfolios tend to outperform in month $t + 1$ when the cross-sectional dispersion in abnormal returns is higher in month t , consistent with more substantial diversification benefits during periods of higher return heterogeneity. In contrast, the β_1 estimate for the value factor is positive but statistically insignificant. Examining the results by subsample, two of three coefficients in the developed sample are statistically insignificant, except for the carry factor, which is significant at the 10% level, suggesting a weaker link between the dispersion in α_k and MV outperformance. However, in the emerging market sample, all three β_1 coefficients are positive and statistically significant at the 10% level or better. These findings suggest that in emerging markets, greater abnormal return dispersion in month t is associated with improved MV portfolio performance in the subsequent month, providing support for the diversification-based explanation in these currencies. Consequently, the evidence rejects Hypothesis 1 (*H1*) in three samples for the carry factor, in all and emerging for the momentum factor, and in the emerging only for the value factor.

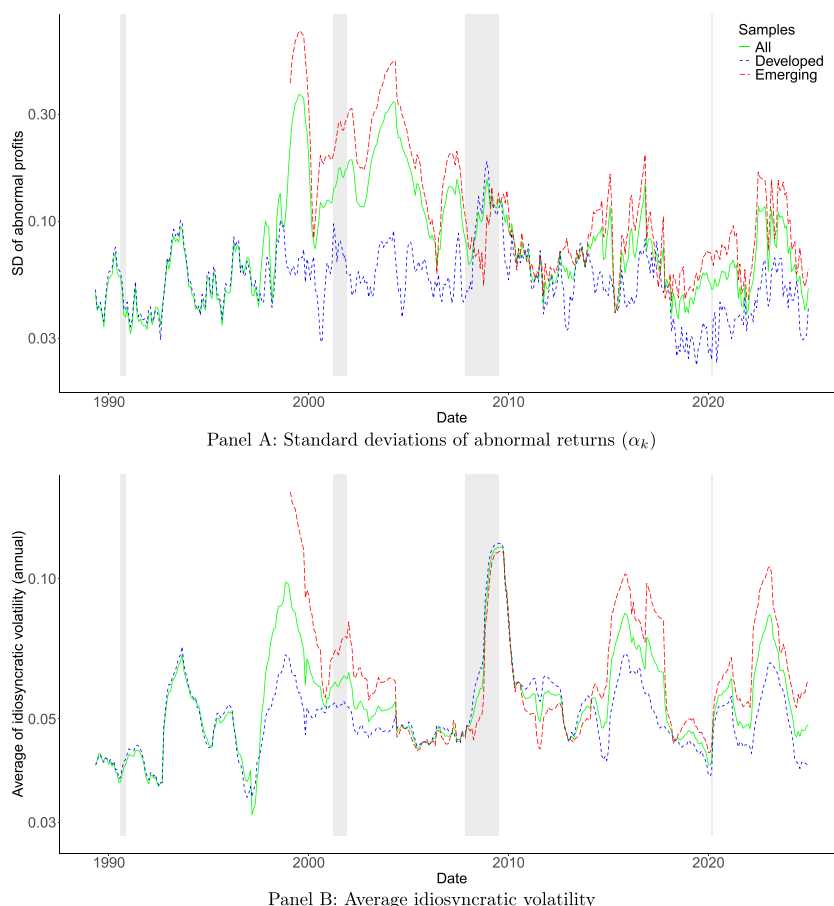


Fig. 5. Time-varying standard deviations of abnormal returns and average idiosyncratic volatility.

This figure plots the time dynamic analysis of abnormal returns and idiosyncratic volatility (IVOL). The y-axis is logarithmically scaled. The solid green line represents the full sample, while the blue and red lines refer to the developed and emerging samples, respectively. Panel A displays the time-dynamic standard deviations of abnormal returns from April 1989 to December 2024. Panel B shows the annualised average idiosyncratic volatility over the same period. The shaded areas represent the recession periods identified by the NBER.

The coefficients on β_2 , which capture the direct effect of average IVOL, are mostly statistically insignificant across all specifications, with one exception: the momentum factor in the developed market sample exhibits a significantly negative coefficient at the 5% level. This isolated result implies that, outside of this case, average IVOL does not materially influence the performance of MV portfolios, leading to a general failure to reject Hypothesis 2 (H_2).

Finally, the interaction term β_3 is significantly negative for all three factors in the emerging sample, further emphasising that the joint effect of high abnormal return dispersion and high IVOL tends to dampen MV portfolio performance. This suggests that although high dispersion of abnormal returns promotes MV outperformance, this effect is moderated when IVOL is also high, particularly in emerging currencies.

Overall, the regression results in Table 5 highlight that the dispersion of abnormal returns is a more critical driver of MV portfolio outperformance than average idiosyncratic volatility, especially in emerging markets. This finding is consistent with the view that international diversification is most effective when return comovements are low and idiosyncratic return variation is high but uncorrelated (Karolyi and Stulz, 1996). In particular, the significant β_1 coefficients in the emerging sample suggest that MV portfolios perform better when abnormal returns are more dispersed, allowing the strategy to overweight currencies with stronger predictive signals. This mechanism is aligned with the theoretical expectation that global diversification benefits arise when market segmentation or structural heterogeneity produces return differences across regions (Bekaert and Harvey, 1995).

5. Optimised currency factors in asset pricing

Introducing the MV-optimised factors raises a new question about whether they can replace the extant currency factors to better explain the cross-section of currency excess returns. First, to investigate whether the optimised factors subsume the abnormal return of the naïve factors, we run spanning tests to assess the relationship following Barillas and Shanken (2017), Ehsani and Linnainmaa (2022), and Fan et al. (2022b). In Table 6, Models (1) to (3) employ the returns of the optimised factors as the dependent variables,

Table 5
Relationship between the outperformance of the MV factor portfolios and the standard deviation of abnormal returns.

Samples		Carry	Mom	Value
All	β_1	2.12**	1.51*	1.44
	t	(2.50)	(1.65)	(1.51)
	β_2	0.11	0.01	-0.09
	t	(1.03)	(0.10)	(-0.98)
	β_3	-30.34**	-23.31	-16.29
	t	(-2.30)	(-1.53)	(-1.19)
Developed	β_1	3.91*	1.03	1.23
	t	(1.93)	(0.48)	(0.57)
	β_2	-0.13	-0.45**	-0.20
	t	(-0.82)	(-2.07)	(-1.10)
	β_3	-24.16	25.93	-13.00
	t	(-0.78)	(0.70)	(-0.38)
Emerging	β_1	0.70**	0.48*	0.71**
	t	(2.52)	(1.83)	(2.16)
	β_2	0.05	0.01	-0.03
	t	(0.77)	(0.19)	(-0.56)
	β_3	-6.74***	-5.51**	-4.98*
	t	(-2.64)	(-2.32)	(-1.83)

This table reports the regression results of Eq. (17). For each factor, we report the correlation coefficients β_1 , β_2 , β_3 , representing the individual impact of the SD of abnormal returns, that of the average IVOL, and the joint effect of these two terms. The corresponding t -values are reported in parentheses, which are measured using the heteroskedasticity- and autocorrelation-consistent (HAC) standard errors proposed by Newey and West (1987). ‘*’, ‘**’, ‘***’ indicate that the t -values of coefficients are statistically significant at the 10%, 5% and 1% levels, respectively.

whereas Models (4) to (6) use the returns of the naïve factors as the dependent variables. We further adopt the US dollar factor of Lustig et al. (2011), the economic and inflation momentum strategies of Dahlquist and Hasseltoft (2020), the change in the VIX index, the TED spread, and foreign exchange volatility as the market controls, and the change in the geopolitical index (GPR) of Caldara and Iacoviello (2022) as macroeconomic controls.

Results in Table 6 reveal that all three optimised strategies generate significant abnormal returns at the 10% level (α of 0.271, 0.263, and 0.200, with t -statistics of 2.86, 2.48, and 1.92, respectively) after controlling for their naïve counterparts and other risk factors. Conversely, when the naïve factors are employed as the dependent variables, their α values are statistically insignificant at the 10% level, with t -statistics of 1.51, -0.26, and 0.27 for carry, momentum, and value, respectively, indicating that their abnormal returns can be fully explained by the optimised counterparts.

When introducing new factors, it is essential to examine their performance within a multi-factor framework.¹³ We follow the methodology of Fama and French (2015). Specifically, we construct 48 test portfolios that proxy for currency risk premia and capture the systematic risks in currency markets. The test portfolios also allow us to assess how the anomalies interact with or reinforce each other. We then estimate a series of multi-factor models, with and without the MV-optimised factors, to evaluate their contributions in explaining the currency risk premia.

Table 7 presents the average annualised excess returns for 4×4 double-sorted portfolios formed by sorting currencies based on any two of the three currency factors.¹⁴ The first currency characteristic is displayed in rows, and the second currency characteristic is shown in columns. For example, Panel A reports the Carry-Momentum 4×4 portfolio returns. Then, different rows represent Q1 (low) to Q4 (high) portfolios sorted on carry signals, and different columns represent Q1 (low) to Q4 (high) portfolios sorted on momentum signals.

In Panel A, momentum works best among high-carry currencies, with 6.69% annual average return observed. This indicates a distinct interaction effect between the two factors that amplifies returns. More specifically, it is observed that the momentum portfolios exhibit stronger performance in the high-carry (Q4) group, with high-minus-low (Q4-Q1) profits reaching 5.53%, compared to the return of 1.16% in the low-carry group. Next, Panels B and C of Table 7 report the results of independent sorts based on carry-value and momentum-value, respectively. The carry and momentum high-minus-low portfolios yield profits across all value groups,

¹³ The widespread availability of advanced computational technologies and AI tools has substantially reduced the costs and complexity of implementing quantitative investment strategies, making portfolio optimisation techniques accessible to a broad range of investors.

¹⁴ Fama and French (2015) formed 5×5 double-sorted portfolios in the US stock markets. Due to the limited number of assets, we adopt the 4×4 double-sorted portfolios to ensure diversification.

Table 6
Spanning test.

Factor	Dependent variable					
	Optimised factors			Naïve factors		
	Carry	Momentum	Value	Carry	Momentum	Value
Model	(1)	(2)	(3)	(4)	(5)	(6)
α (%)	0.271*** (2.86)	0.263** (2.48)	0.200* (1.92)	0.098 (1.51)	−0.019 (−0.26)	0.02 (0.27)
Opt				0.297*** (6.54)	0.386*** (11.1)	0.254*** (6.65)
Dollar	0.027 (0.36)	−0.076 (−0.93)	−0.133 (−1.43)	0.124** (2.27)	0.042 (0.79)	0.1 (1.56)
Carry	0.680*** (7.81)	0.021 (0.23)	0.06 (0.64)		0.041 (0.65)	−0.008 (−0.09)
MOM	0.402*** (6.36)	1.092*** (14.04)	0.443*** (5.60)	−0.071 (−1.09)		−0.274*** (−5.08)
Value	−0.260** (−2.59)	−0.258*** (−2.60)	0.691*** (7.41)	0.084 (1.05)	−0.063 (−0.99)	
EconMOM	0.026 (0.52)	0.026 (0.43)	0.048 (1.03)	−0.004 (−0.15)	−0.015 (−0.38)	−0.035 (−1.22)
InfMOM	0.004 (0.09)	−0.034 (−0.48)	−0.014 (−0.27)	0.001 (0.05)	0.02 (0.47)	0.026 (0.76)
Δ GPR	−0.001 (−0.22)	0.003 (0.76)	0.001 (0.00)	0.001 (0.27)	−0.003 (−1.29)	−0.004* (−1.65)
Δ VIX	0.001 (−0.06)	0.001 (0.39)	0.001 (−0.74)	−0.001** (−2.41)	0.001 (0.97)	0.001 (−0.58)
Δ TED	0.001 (0.09)	0.001 (0.89)	0.001 (0.41)	−0.001* (−1.77)	−0.001** (−2.09)	0.001 (0.41)
Δ FXVOL	−0.001 (−1.32)	−0.001 (−0.94)	0.001 (−0.23)	−0.001 (−1.20)	0.002** (2.51)	0.001 (0.29)

This table presents the results of spanning tests for the MV-optimised factors and the naïve ones, respectively, as the dependent variables. α is in percentage. Dollar denotes the dollar factor of [Lustig et al. \(2011\)](#). EconMOM and InfMOM denote the economic and inflation momentum factors of [Dahlquist and Hasseltoft \(2020\)](#), respectively. Δ GPR refers to the changes in the geopolitical risk index of [Caldara and Iacoviello \(2022\)](#). Δ VIX, Δ TED, and Δ FXVOL refer to changes in the VIX volatility index, the TED spread, and foreign exchange volatility (FXVOL), respectively, all of which are non-traded factors. For each model, we report the factor loadings and the corresponding t -values. The t -value is measured by the heteroskedasticity- and autocorrelation-consistent (HAC) standard errors proposed by [Newey and West \(1987\)](#). ‘*’, ‘**’, ‘***’ represent that the t -values of coefficients are statistically significant at the 10%, 5% and 1% levels, respectively. The sample period ranges from April 1989 to December 2024.

with increasing profits from low-carry/momentum to high-carry/momentum groups. This highlights the robustness and persistence of these two effects in the currency market. However, the value effect appears weak. Specifically, although Q4-minus-Q1 value portfolios generate meaningful profits, the value profits do not continue to grow from low-value to high-value groups. These results are consistent with the low profitability of the value factor observed in [Table 1](#).

We next evaluate the ability of the optimised factors to explain average excess returns on the test portfolios, as reported in [Table 8](#). In Panel A, we include the dollar factor of [Lustig et al. \(2011\)](#) to construct factor models. For robustness, Panel B incorporates the inflation and economic momentum factors of [Dahlquist and Hasseltoft \(2020\)](#) in addition to the dollar factor as controls. The benchmark model in both panels is Model (1), which includes the three naïve factors—carry, momentum, and value.

Under the null hypothesis of a well-specified asset pricing model, the regression intercepts of excess returns on factor returns should equal zero ([Fama and French, 1992](#)). A statistically significant intercept implies the existence of abnormal returns that are unexplained by the model’s risk factors, suggesting an incomplete spanning of the return-generating process. We apply the GRS statistic proposed by [Gibbons et al. \(1989\)](#) on combinations of naïve and optimised factors to test this hypothesis.

We first assess the benchmark asset pricing model (Model 1), which includes the three naïve currency factors: carry, momentum, and value. This model yields a GRS statistic of 1.331 with a corresponding p -value of 0.078. Applying a 10% significance threshold,

Table 7

Average annual excess returns for portfolios formed on carry and momentum, carry and value, momentum and value.

	Annual average return				t-values			
	Q1	Q2	Q3	Q4	Q1	Q2	Q3	Q4
Panel A: Carry-Momentum portfolios								
Q1	−1.92%	−1.10%	−1.12%	−0.76%	−1.42	−0.81	−1.03	−0.70
Q2	0.78%	−0.47%	1.50%	−0.31%	0.58	−0.36	1.19	−0.23
Q3	0.49%	−1.08%	1.65%	1.03%	0.34	−0.74	1.16	0.66
Q4	1.16%	−0.20%	1.60%	6.69%	0.48	−0.13	1.08	4.04
Panel B: Carry-Value portfolios								
Q1	−1.48%	−1.62%	−1.69%	−1.11%	−1.25	−1.37	−1.41	−0.96
Q2	−0.93%	−0.14%	1.21%	0.73%	−0.68	−0.11	0.83	0.63
Q3	−0.50%	1.94%	0.92%	0.58%	−0.38	1.51	0.62	0.37
Q4	2.44%	3.03%	2.31%	4.07%	1.57	1.72	1.15	2.06
Panel C: Momentum-Value portfolios								
Q1	−2.18%	0.74%	−0.41%	0.08%	−1.64	0.41	−0.27	0.05
Q2	−0.02%	−0.81%	0.38%	0.86%	−0.01	−0.67	0.26	0.64
Q3	1.17%	1.17%	−0.01%	2.24%	0.93	0.96	−0.01	1.57
Q4	2.99%	2.43%	1.82%	4.84%	2.13	1.44	1.32	3.02

This table reports the annualised profits of the test portfolios based on an approach by Fama and French (2015). We employ three currency characteristics in various combinations to form the portfolios. Panel A presents Carry-Momentum portfolios. More specifically, currencies are first allocated across four rows based on their carry signals (Q1 to Q4, representing low to high carry). Second, currencies are allocated independently across four columns based on their momentum signals (Q1 to Q4, representing low to high momentum), using the past three excess returns as breakpoints. The intersections of the two sorts produce 16 equally weighted Carry-Momentum portfolios. The Carry-Value (Panel B) and Momentum-Value (Panel C) portfolios are formed in the same way. The t-value calculation uses heteroskedasticity- and autocorrelation-consistent (HAC) standard errors as proposed by Newey and West (1987). The sample period ranges from April 1989 to December 2024.

Table 8

Summary statistics for multi-factor model test.

Model		GRS	p-value	$A \alpha $	$\frac{A \alpha }{A \bar{r}_i }$	$\frac{A(\alpha^2)}{A(\bar{r}_i^2)}$	$A(R^2)$
Panel A: Dollar factor as an additional factor							
(1)	Carry + MOM + Value	1.331	0.078	7.34%	0.644	0.337	57.6%
(2)	Opt.Carry + MOM + Value	1.371	0.058	8.07%	0.708	0.402	55.4%
(3)	Carry + Opt.MOM + Value	1.118	0.282	7.19%	0.631	0.308	55.5%
(4)	Carry + MOM + Opt.Value	1.317	0.085	7.29%	0.639	0.320	55.4%
Panel B: Dollar factor with the economic and inflation momentum as additional factors							
(1)	Carry + MOM + Value	1.332	0.077	7.35%	0.645	0.337	57.9%
(2)	Opt.Carry + MOM + Value	1.375	0.057	8.10%	0.710	0.405	55.7%
(3)	Carry + Opt.MOM + Value	1.122	0.276	7.20%	0.632	0.310	55.7%
(4)	Carry + MOM + Opt.Value	1.321	0.083	7.30%	0.641	0.323	55.6%

This table evaluates the performance of multi-factor models in explaining the monthly excess returns of the 48 test portfolios detailed in Table 7. Panel A presents results where the factors augment the dollar factor, while Panel B extends the analysis by including the dollar factor along with economic and inflation momentum factors from Dahlquist and Hasseltoft (2020). We construct the dollar factor across all 48 currencies in our sample following Lustig et al. (2011). The GRS statistic tests whether the expected values of the 48 intercept estimates are jointly zero, with the corresponding p-values reported. Key measures include $A|\alpha|$, $\frac{A|\alpha|}{A|\bar{r}_i|}$, $\frac{A(\alpha^2)}{A(\bar{r}_i^2)}$, and $A(R^2)$. Specifically, $A|\alpha|$ represents the average absolute value of the intercepts, $\frac{A|\alpha|}{A|\bar{r}_i|}$ captures the ratio of the average absolute intercept to the average absolute value of $\bar{r}_i$, and $\frac{A(\alpha^2)}{A(\bar{r}_i^2)}$ measures the ratio of the average squared intercept to the average squared value of $\bar{r}_i$. Here, $\bar{r}_i$ denotes the average return of portfolio i adjusted by subtracting the mean return across all test portfolios. $A(R^2)$ refers to the average R^2 across all regressions.

this p-value leads to a rejection of the null hypothesis that the model's intercept estimates (alphas) are jointly zero. This result is a key finding, as it provides clear statistical evidence that the standard three-factor model composed entirely of naïve currency factors is insufficient to explain the cross-section of currency returns.

Our next focus is on the change in model performance when replacing the naïve factors with their optimised counterparts. This procedure is grounded in both economic rationale. Economically, although MV optimisation aims to enhance return efficiency by

mitigating unrewarded risk exposures, the naïve factors may still capture distinct systematic information that is not fully represented in the optimised versions. This is particularly relevant for momentum and value, where decomposition results in Table 4 reveal significant differences in return components, such as mean, volatility, and risk-adjusted performance, between the naïve and MV strategies.

Replacing the naïve carry or value factor, Model (2) and (4), increases the GRS statistics, suggesting no improvement in these cases. However, replacing the naïve momentum factor with the optimised factors can reduce the GRS statistic to 1.118 with p -value at 0.282, which does not reject the null hypothesis at any level. This indicates that its pricing errors are statistically indistinguishable from zero, demonstrating that this combination of naïve and optimised factors successfully explains the average excess returns of the 48 test portfolios where the other models failed. Model (3) results demonstrate that the combined use of naïve and MV factors can improve the model's explanatory power, supporting the idea that combining the naïve and optimised factors offers complementary insights in building the asset pricing model. With the dollar factor, we refer to Model (3) as the new four-factor model in the following discussion.

Table 8 also reports two ratios introduced by Fama and French (2015) that estimate the proportion of the cross-section of expected returns left unexplained by competing models. The numerator of each measure reflects the dispersion of intercepts produced by a given model for a set of test portfolios, while the denominator captures the dispersion of test portfolios' expected returns. Let R_i represent the time series average excess returns for portfolio i , $\bar{R}$ denote the cross-sectional average of R_i , and r_i be portfolio i 's deviation from the cross-sectional average, defined as $r_i = R_i - \bar{R}$. The first estimate is expressed as $\frac{A(|\alpha_i|)}{A(|r_i|)}$, which is the average absolute value of the intercepts, $|\alpha_i|$, divided by the average absolute value of r_i . This metric evaluates the proportion of the dispersion in intercepts relative to the dispersion in portfolio deviations from the cross-sectional average return.

The results for $\frac{A(|\alpha_i|)}{A(|r_i|)}$ in Table 8 show that, for various test portfolios, the benchmark model's absolute intercept, $A(|\alpha_i|)$, accounts for 0.644 of $A(|r_i|)$. This finding indicates that the benchmark model fails to explain 64.4% of the dispersion in average excess returns, as measured in units of return. When replacing carry or value naïve factor with the optimised versions, the proportion of unexplained dispersion in average excess returns ranges from 0.639 to 0.708. Moving to the new four-factor model, Model (3) achieves a remarkable improvement by reducing the ratio to 0.631. This result implies that only 63.1% of the average excess returns remain unexplained, representing the best performance across all models.

The second estimate captures the extent of error inflation by measuring increases in both the average absolute intercept, $A(|\alpha_i|)$, and the average absolute deviation, $A(|r_i|)$. The estimated intercept, a_i , is the sum of the true intercept, α_i , and an estimation error, e_i , expressed as $a_i = \alpha_i + e_i$. Similarly, r_i , the deviation of portfolio i 's return from the cross-sectional average, is the sum of μ_i , portfolio i 's expected deviation from the grand mean, and an estimation error, ϵ_i , such that $r_i = \mu_i + \epsilon_i$. To account for the impact of measurement error, we focus on squared intercepts and squared deviations, which mitigate the inflation effects introduced by the errors and provide a more accurate adjustment. The cross-sectional average of μ_i is zero, so $A(\mu_i^2)$ represents the cross-sectional variance of expected portfolio returns, and the ratio $A(\alpha_i^2)/A(\mu_i^2)$ quantifies the proportion of $A(\mu_i^2)$ left unexplained by a model. Since α_i is a constant, the expected value of the squared estimated intercept is equal to the squared value of the true intercept plus the sampling variance of the estimate, $E(a_i^2) = \alpha_i^2 + E(e_i^2)$. Our estimate, $\hat{\alpha}_i^2$, of the squared true intercept, α_i^2 , is calculated as the difference between the squared value of the estimated regression intercept and the square of its standard error. Similarly, our estimate of μ_i^2 , $\hat{\mu}_i^2$, is derived as the difference between the squared realised deviation, r_i^2 , and the square of its standard error. The ratio of averages, $A(\hat{\alpha}_i^2)/A(\hat{\mu}_i^2)$, then provides an estimate of the proportion of the variance of the expected returns of the test portfolio that remains unexplained by the model.

As highlighted in Fama and French (2015), compared to $A(|\alpha_i|)$, the ratio $A(\hat{\alpha}_i^2)/A(\hat{\mu}_i^2)$ is expressed in units of squared return and accounts for corrections due to sampling error. The estimates indicate that the benchmark model leaves 33.7% of the cross-sectional variance of expected returns unexplained. Notably, in our four-factor model (Model 3), this ratio is reduced to just 0.308, indicating that only 30.8% of the cross-sectional variance of expected returns remains unexplained.

The superior performance of Model (3) likely arises from the distinct way in which optimisation refines the currency momentum signal relative to the other factors. As shown in Table 4, the MV momentum factor enhances the capture of cross-sectional return dynamics (Components 2 and 3), whereas the naïve momentum factor relies primarily on time-series predictability (Component 1). Notably, the diversification-benefit contribution from Component (2) improves substantially, shifting from -11.85% for the naïve momentum factor to 1.04% for its MV-optimised counterpart.

Currency momentum reflects the persistence of recent exchange rate trends, often linked to information diffusion or extrapolative expectations in FX markets. However, the naïve equal-weighted construction may inadvertently embed noise or overlapping exposures to systematic risks already captured by the carry or value factors. Through shrinkage-based optimisation (Jorion, 1986), the MV framework mitigates these redundant exposures, yielding a cleaner and more distinctive trend component. Consequently, the MV momentum factor is likely more orthogonal to the carry and value factors, thereby enhancing its incremental explanatory power. This complementarity, consistent with evidence that diversified FX style combinations capture distinct risk premia (Kroencke et al., 2014), offers an economic rationale for the superior GRS statistic and the inability to reject the null hypothesis in Model (3).

In conclusion, the results for $\frac{A(|\alpha_i|)}{A(|r_i|)}$ underscore the effectiveness of incorporating optimised factors into the model, as they significantly enhance its ability to explain the dispersion of average excess returns, thereby improving the model's overall explanatory power. Similarly, the findings for $A(\hat{\alpha}_i^2)/A(\hat{\mu}_i^2)$ further highlight the substantial gains in explanatory power achieved through the inclusion of optimised factors. These results demonstrate the superiority of the six-factor model in capturing cross-sectional variations in expected returns, reinforcing its robustness and effectiveness in asset pricing applications.

Panel B of Table 8 presents the robustness check after incorporating economic and inflation momentum factors from Dahlquist and Hasseltoft (2020) as additional macroeconomic level factors. The results substantiate the superior performance of Model (3) relative to alternative specifications. Specifically, when augmented with two additional macroeconomic control variables, Model (3) exhibits the most favourable pricing performance across multiple statistical metrics. The model yields the lowest GRS statistic of 1.123 ($p = 0.276$), the only one that fails to reject the null hypothesis of zero alphas at the 10% level. Furthermore, Model (3) demonstrates the most substantial pricing efficiency as evidenced by both the lowest $\frac{A(|\alpha_i|)}{A(|r_i|)}$ ratio of 0.632 and the lowest $A(\hat{\alpha}_i^2)/A(\hat{\mu}_i^2)$ value of 0.310. These results provide robust evidence that using MV-optimised factors in addition to naïve factors enhances the model's ability to capture the cross-sectional variation of currency returns.

6. Robustness checks

This section provides a comprehensive set of robustness checks for the outperformance of our MV-optimised factors. We evaluate whether the baseline results are sensitive to key design choices by examining three dimensions. First, because our baseline adopts a 50/50 high-minus-low sorting scheme that differs from Menkhoff et al. (2012a), we re-estimate factor performance using their portfolio construction to facilitate comparability with the benchmark literature. Second, given that a range of covariance estimators has been developed beyond the Bayesian-Stein framework, we consider alternative estimators to ensure that the outperformance is not driven by a specific covariance specification. Third, to address potential concerns about liquidity in a broad currency universe, we replicate the analysis using an alternative sample concentrated on more liquid currencies. Overall, the results remain consistent with our main findings and continue to support the superior performance of the MV factors.

6.1. Alternative high-minus-low portfolio set

To assess robustness and comparability with the existing literature, we perform an analysis using the widely employed tercile-sorted portfolio construction, as in Lustig et al. (2011) and Menkhoff et al. (2012a). While the primary 50/50 split represents a stringent test of the MV framework due to its broad cross-sectional coverage, this robustness exercise verifies that the superior performance of the MV factors is not driven by the specific portfolio design.

Table 9 reports the OOS performance of these tercile-sorted portfolios. The results confirm the findings in Table 1, showing that the MV-optimised factors consistently and significantly outperform their naïve counterparts. The mean annual return differentials are both economically and statistically significant across all three strategies: the MV carry factor by 1.714% ($t = 1.736$), the momentum factor outperforms by 3.198% ($t = 2.911$), and the value factor by 1.852% ($t = 1.775$). Improvements in risk-adjusted performance are also robust. The bootstrapped Sharpe ratio equality test of Ledoit and Wolf (2008) indicates that the Sharpe ratio of each MV factor significantly exceeds that of its naïve counterpart at the 10% level. Consistent with these results, other OOS performance metrics, including the Omega ratio, Dowd ratio, and CER, support the outperformance of the MV factors.

6.2. Alternative covariance estimators

Our main analysis intentionally employs a conventional sample-based covariance matrix to provide a conservative estimate of the MV framework's benefits. To test the robustness of our findings and examine the impact of more advanced techniques, we next re-evaluate the performance of the MV factors using several alternative covariance estimators prominent in the literature. We applied three estimation approaches: the Exponentially Weighted Moving Average (EWMA), the shrinkage estimator of Ledoit and Wolf (2003), and the oracle approximating shrinkage estimator of Chen et al. (2010). Such models are widely used and can serve as efficient proxies for covariance estimators over a long period. For a more comprehensive analysis of adopting various covariance estimators, please refer to Fan et al. (2025).

Table 9

Out-of-sample performance using tercile sorted MV-optimised factors.

Strategies	Carry			Momentum			Value		
	naïve	MV	Diff	naïve	MV	Diff	naïve	MV	Diff
Mean	4.492%***	6.206%***	1.714%*	2.010%**	5.208%***	3.198%***	1.248%	3.099%***	1.852%*
t-value	(4.018)	(4.614)	(1.736)	(2.012)	(3.868)	(2.911)	(1.270)	(2.486)	(1.775)
Vol	0.061	0.076	0.015	0.058	0.093	0.034	0.057	0.069	0.012
SR	0.733***	0.816***	0.084*	0.346**	0.562***	0.216***	0.219	0.449***	0.230*
SR.t	(4.195)	(4.649)	(1.681)	(2.084)	(3.288)	(2.652)	(1.310)	(2.692)	(1.760)
Omega	1.736	1.880	0.144	1.312	1.572	0.260	1.177	1.408	0.231
Dowd	0.136	0.158	0.023	0.072	0.113	0.040	0.041	0.091	0.050
CER(%)	0.296%	0.397%	0.101%	0.097%	0.255%	0.158%	0.037%	0.159%	0.123%

This table summarises the out-of-sample performance of the naïve and mean-variance factor portfolios with transaction costs from April 1989 to December 2024 based on long the top 1/3 high signal currencies and short the bottom low signal currencies. This portfolio set aligns with the existing literature. Diff refers to the differences between naïve and mean-variance optimised factors' performance. Mean, t-value, Vol, and Sharpe denote the annualised average returns, standard deviation, t-statistic of mean, and Sharpe ratio of portfolio returns. The t-value (in parentheses) uses heteroskedasticity- and autocorrelation-consistent (HAC) standard errors proposed by Newey and West (1987). SR.t denotes the t-value of the Sharpe ratio based on the bootstrapped Sharpe ratio equality test of Ledoit and Wolf (2008). Omega, Dowd, and CER refer to the Omega ratio, Dowd ratio, and Certainty Equivalent Return, respectively. *, **, *** represent that the t-values are statistically significant at 10%, 5%, and 1% levels.

Table 10

Out-of-sample performance using naïve and mean-variance optimised factors (alternative covariance estimators).

Strategies	Carry		Momentum		Value	
	MV	Diff	MV	Diff	MV	Diff
Panel A: exponential weighted moving average (EWMA) estimator						
<i>Mean</i>	5.641%***	2.430%**	4.222%***	2.479%**	3.405%***	2.556%**
<i>t</i> -value	(4.21)	(2.16)	(3.92)	(2.13)	(2.78)	(2.22)
<i>Vol</i>	0.077	0.031	0.093	0.040	0.071	0.026
<i>SR</i>	0.729***	0.038**	0.453***	0.124**	0.480***	0.291**
<i>SR.t</i>	(4.19)	(2.07)	(2.69)	(2.01)	(2.87)	(2.16)
<i>Omega</i>	1.748	0.055	1.425	0.140	1.451	0.300
<i>Dowd</i>	0.139	0.014	0.088	0.023	0.097	0.062
<i>CER</i> (%)	0.345%	0.123%	0.171%	0.084%	0.179%	0.150%
Panel B: Ledoit-Wolf shrinkage estimator						
<i>Mean</i>	5.646%***	2.434%**	4.419%***	2.676%**	3.525%***	2.677%**
<i>t</i> -value	(4.47)	(2.30)	(3.65)	(2.36)	(2.91)	(2.35)
<i>Vol</i>	0.074	0.028	0.092	0.039	0.071	0.026
<i>SR</i>	0.758***	0.067**	0.481***	0.152**	0.497***	0.308**
<i>SR.t</i>	(4.36)	(2.14)	(2.87)	(2.24)	(2.96)	(2.25)
<i>Omega</i>	1.787	0.095	1.460	0.175	1.472	0.321
<i>Dowd</i>	0.147	0.022	0.097	0.032	0.099	0.064
<i>CER</i> (%)	0.355%	0.132%	0.193%	0.106%	0.189%	0.160%
Panel C: Oracle approximating shrinkage estimator						
<i>Mean</i>	5.671%***	2.460%**	4.445%***	2.702%**	3.552%***	2.703%**
<i>t</i> -value	(4.46)	(2.32)	(3.65)	(2.36)	(3.00)	(2.37)
<i>Vol</i>	0.075	0.028	0.092	0.039	0.071	0.026
<i>SR</i>	0.759***	0.068**	0.484***	0.155**	0.501***	0.312**
<i>SR.t</i>	(4.36)	(2.15)	(2.89)	(2.26)	(2.98)	(2.27)
<i>Omega</i>	1.790	0.097	1.463	0.178	1.476	0.325
<i>Dowd</i>	0.147	0.022	0.098	0.033	0.100	0.065
<i>CER</i> (%)	0.356%	0.134%	0.195%	0.108%	0.191%	0.163%

This table summarises the out-of-sample performance of the mean-variance factor portfolios with transaction costs from April 1989 to December 2024. Panels A to C report results based on alternative covariance estimators: the EWMA, the Ledoit–Wolf shrinkage estimator of Ledoit and Wolf (2003), and the oracle approximating shrinkage estimator of Chen et al. (2010), respectively. Diff refers to the differences between naïve and mean-variance optimised factors' performance. *Mean*, *t*-value, *Vol*, and *Sharpe* denote the annualised average returns, standard deviation, *t*-statistic of mean, and Sharpe ratio of portfolio returns. The *t*-value (in parentheses) uses heteroskedasticity- and autocorrelation-consistent (HAC) standard errors proposed by Newey and West (1987). *SR.t* denotes the *t*-value of the Sharpe ratio based on the bootstrapped Sharpe ratio equality test of Ledoit and Wolf (2008). *Omega*, *Dowd*, and *CER* refer to the Omega ratio, Dowd ratio, and Certainty Equivalent Return, respectively. *, **, *** represent that the *t*-values are statistically significant at 10%, 5%, and 1% levels.

Table 11

Out-of-sample performance using naïve and mean-variance optimised factors (sample of Ackermann et al., 2017).

Strategies	Carry			Momentum			Value		
	naïve	MV	Diff	naïve	MV	Diff	naïve	MV	Diff
<i>Mean</i>	3.539%***	5.289%***	1.750%**	0.711%	3.118%***	2.407%***	1.292%	2.740%**	1.448%*
<i>t</i> -value	(3.533)	(4.736)	(2.071)	(0.691)	(2.725)	(3.016)	(1.395)	(2.415)	(1.861)
<i>Vol</i>	0.055	0.072	0.017	0.061	0.074	0.014	0.054	0.066	0.012
<i>SR</i>	0.648	0.735	0.087	0.117	0.420	0.302	0.239	0.416	0.177
<i>SR.t</i>	(3.791)	(4.267)	(2.015)	(0.704)	(2.514)	(2.946)	(1.428)	(2.494)	(1.792)
<i>Omega</i>	1.633	1.753	0.120	1.095	1.385	0.291	1.198	1.367	0.170
<i>Dowd</i>	0.125	0.143	0.018	0.023	0.086	0.063	0.045	0.083	0.038
<i>CER</i> (%)	0.233%	0.333%	0.100%	−0.017%	0.145%	0.162%	0.047%	0.138%	0.091%

This table summarises the out-of-sample performance of the naïve and mean-variance factor portfolios with transaction costs from April 1989 to December 2024 based on the sample of Ackermann et al. (2017). Diff refers to the differences between naïve and mean-variance optimised factors' performance. *Mean*, *t*-value, *Vol*, and *Sharpe* denote the annualised average returns, standard deviation, *t*-statistic of mean, and Sharpe ratio of portfolio returns. The *t*-value (in parentheses) uses heteroskedasticity- and autocorrelation-consistent (HAC) standard errors proposed by Newey and West (1987). *SR.t* denotes the *t*-value of the Sharpe ratio based on the bootstrapped Sharpe ratio equality test of Ledoit and Wolf (2008). *Omega*, *Dowd*, and *CER* refer to the Omega ratio, Dowd ratio, and Certainty Equivalent Return, respectively. *, **, *** represent that the *t*-values are statistically significant at 10%, 5%, and 1% levels.

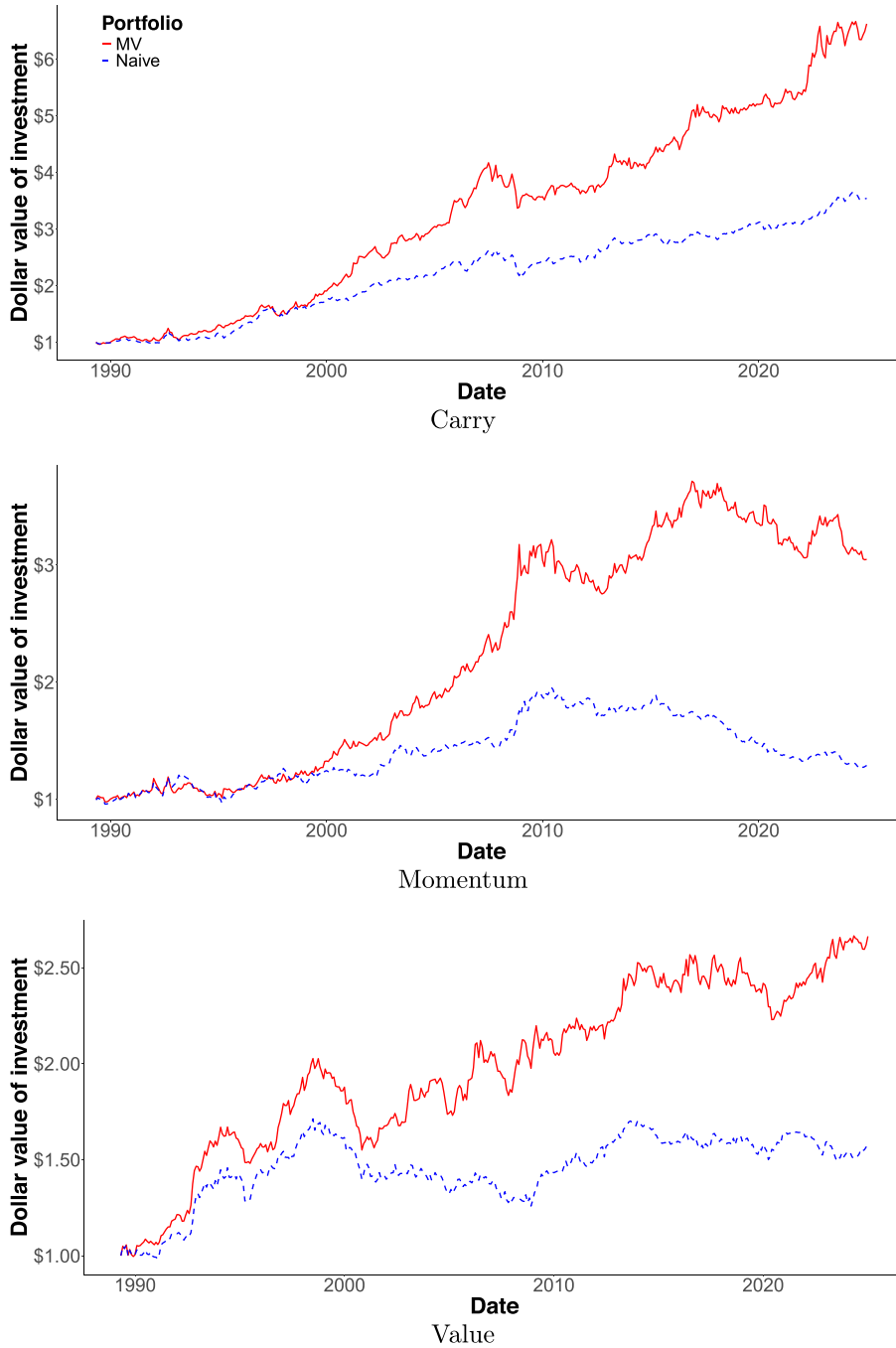


Fig. 6. Cumulative performance of the currency factor portfolios (robustness).

These plots exhibit the cumulative performance of the MV-optimised and the naïve factor portfolios based on the sample of [Ackermann et al. \(2017\)](#). The dollar value of investment (y-axis) is logarithmically scaled, given the considerable difference between the two weighting schemes. MV and naïve refer to the mean-variance diversified factors and the original equally weighted factors, respectively. The sample ranges from April 1989 to December 2024.

The EWMA model is a simple dynamic model in which recent returns have more weight than past returns in the estimation. The out-of-sample covariance matrix is measured as:

$$\Sigma_t = \lambda \hat{\Sigma}_t + (1 - \lambda) \hat{\mu}_t' \hat{\mu}_t, \quad (18)$$

where $\hat{\mu}_t'$ is the transpose of return vector $\hat{\mu}_t$. Following [Ardia and Boudt \(2015\)](#), we determined λ to be 0.94.

The Ledoit-Wolf shrinkage estimator is a weighted average sample covariance matrix based on 'prior' or 'shrinkage target', which was first built by [Ledoit and Wolf \(2003\)](#) and further developed by [Ledoit and Wolf \(2004a,b\)](#). We adopt the approach of

Ledoit and Wolf (2003) and set the ‘shrinkage target’ as given by a single-factor model of Sharpe (1964), where the factor is equal to the cross-sectional average of all the variables.

The oracle approximating shrinkage estimator of Chen et al. (2010) is designed to minimise the mean squared error of covariance estimation. The authors proposed estimating the out-of-sample covariance matrix by iteratively approximating the shrinkage with:

$$\Sigma_t = \rho \hat{F}_t + (1 - \rho) \hat{\Sigma}_t \quad (19)$$

where $\rho \in (0, 1)$ is a control parameter, and $\hat{F}_t$ is a target matrix. It is proposed to use a structured estimate $\hat{F}_t = \text{Tr}(\hat{\Sigma}_t/p) \cdot I_{p \times p}$ where $I_{p \times p}$ is an identity matrix of dimension p , and $\text{Tr}(\bullet)$ represents the trace of a matrix. In our study, dimension p is equal to the length of the estimation window, M . The control parameter is estimated as:

$$\rho = \min\left(\frac{(\frac{1-2}{p})\text{Tr}(\hat{\Sigma}^2) + \text{Tr}^2(\hat{\Sigma})}{(\frac{N+1-2}{p})[\text{Tr}(\hat{\Sigma}^2) - \frac{\text{Tr}^2(\hat{\Sigma})}{p}]}, 1\right), \quad (20)$$

where all notations remain the same.

Table 10 reports the out-of-sample (OOS) performance of the MV factors constructed using alternative covariance estimators: the Exponentially Weighted Moving Average (EWMA) estimator (Panel A), the Ledoit–Wolf shrinkage estimator (Panel B), and the oracle approximating shrinkage estimator (Panel C). The results indicate that the outperformance of the MV factors from Table 1 is robust. Across all panels and factors, the improvements in mean returns remain statistically significant at the 5% level. The bootstrapped Sharpe ratio equality test of Ledoit and Wolf (2008) further confirms that the risk-adjusted outperformance is statistically significant at the 5% level in all cases. Other OOS performance metrics, including the Omega ratio, Dowd ratio, and CER, consistently support our main text findings.

We also observe a monotonic improvement in both profitability and risk-adjusted performance as the covariance estimator is published more recently. The oracle approximating shrinkage estimator in Panel C delivers the highest mean returns and Sharpe ratios, suggesting that although our main results are robust, the economic gains from MV optimisation can be further enhanced through more accurate covariance estimation.

Three economic implications follow from this estimator ranking. First, the shrinkage estimators of Ledoit and Wolf (2003) and Chen et al. (2010) consistently deliver the highest annualised returns and Sharpe ratios across all three factors. The oracle approximating shrinkage (OAS) estimator raises the Sharpe ratios of the optimised carry, momentum, and value factors to 0.759, 0.484, and 0.501, respectively, marginally above both the Bayes–Stein baseline and the Ledoit–Wolf estimator. This ordering is consistent with the prediction of Ledoit and Wolf (2003) and DeMiguel et al. (2009a) that structured shrinkage reduces covariance estimation error and yields more efficient out-of-sample portfolio weights. Second, the EWMA estimator, which imposes no structured shrinkage, delivers the weakest performance of the three alternatives, though it still significantly outperforms the naïve factors; the Sharpe ratio gain from switching to a structured shrinkage method is approximately 0.02–0.03 across factors, an economically meaningful difference over the full sample. Third, the incremental gain from Ledoit–Wolf or OAS relative to the Bayes–Stein baseline is modest, confirming that the outperformance of the MV factors is not specific to any particular shrinkage estimator. Taken together, shrinkage clearly dominates non-shrinkage estimators in currency MV optimisation, while differences within the shrinkage family are of second-order importance.

6.3. Empirical results based on a smaller and more liquid sample

As the last robustness check, we re-estimate the MV-optimised currency portfolios using a more liquid and actively traded sample drawn from Ackermann et al. (2017), which includes 19 actively traded currencies. This alternative sample comprises the Australian Dollar, Canadian Dollar, Czech Koruna, Euro, Hungarian Forint, Icelandic Krona, Israeli Shekel, Japanese Yen, Mexican Peso, New Zealand Dollar, Norwegian Krone, Polish Zloty, Russian Ruble, Singapore Dollar, South African Rand, Swedish Krona, Swiss Franc, Thai Baht, and UK Pound.¹⁵ The reduced set allows us to assess whether the performance of the MV strategy is sensitive to the inclusion of less liquid or more volatile currencies, and to verify that our main findings are not driven by estimation noise introduced by a broader and potentially less stable currency universe.

Table 11 reports the out-of-sample performance of the naïve and MV-optimised factor portfolios using the alternative sample from Ackermann et al. (2017). Consistent with the results in Table 1, the carry factor delivers the strongest performance among the naïve portfolios, yielding an average annual return of 3.539% and a Sharpe ratio of 0.648. The value factor ranks second, with a return of 1.292% and a Sharpe ratio of 0.239. The momentum factor becomes weak and produces a statistically insignificant return of 0.711% under the naïve specification. The carry strategy also dominates across alternative performance measures, including the Omega ratio (1.633), Dowd ratio (0.125), and certainty-equivalent return (CER) of 0.233%. Overall, naïve factor strategies based on this smaller, more liquid sample underperform relative to the results from our large sample in the main body of the paper, reflecting more limited cross-sectional dispersion and lower signal strength.

Importantly, incorporating MV optimisation materially enhances portfolio performance in the robustness sample as well. As shown in Table 11, the MV-optimised portfolios generate average returns at least 1.448% higher than their naïve counterparts. While the magnitude of improvement is somewhat smaller than in the main sample, the return differentials remain statistically significant at

¹⁵ Due to data limitations, we exclude the Turkish Lira from our sample.

the 10% level. The momentum factor exhibits the largest performance gain, with an increase of 2.407 percentage points in annual return following MV optimisation. In addition, *t*-statistics on the Sharpe ratios confirm that the improvements in risk-adjusted returns for all three factors are significant at the 10% level. These findings suggest that the benefits of return-based portfolio optimisation are robust to changes in currency universe composition and persist across different samples.

Fig. 6 presents the cumulative return performance of the MV-optimised and naïve factor portfolios using the reduced currency set from Ackermann et al. (2017). The performance patterns are broadly consistent with those observed in the main sample (Fig. 1). Across all three strategies, the MV-optimised portfolios (depicted in red) generate substantially higher cumulative returns than their naïve counterparts, in line with the summary statistics reported in Table 11.

For the carry and momentum factors, the cumulative performance differentials are modest prior to 2000, reflecting relatively muted benefits from optimisation during this period. However, the divergence becomes increasingly pronounced in the post-2000 era, with MV-optimised portfolios exhibiting consistently stronger performance. In the case of the value factor, the MV-optimised portfolio outperforms its naïve counterpart throughout the entire sample period, including the pre-2000 window. These results reinforce the robustness of the MV approach in enhancing long-run performance across different samples and time horizons.

This robustness analysis, based on a more liquid currency sample from Ackermann et al. (2017), reinforces the main empirical findings. Although the naïve factor portfolios exhibit weaker baseline performance in the reduced sample, mean-variance optimisation consistently enhances both returns and Sharpe ratios across all three factors. The value factor, which initially delivers negative excess returns, becomes profitable following optimisation, while the cumulative return patterns remain broadly consistent with those observed in the main sample. These results confirm that the performance gains associated with MV optimisation are not driven by the breadth of the currency universe or the inclusion of illiquid assets, but instead reflect economically meaningful improvements in portfolio construction.

7. Conclusion

This study proposes the application of MV optimisation on extant currency factors. By adopting a Bayesian mean-variance portfolio construction, we find compelling evidence to support the superiority of the MV-optimised factors over the simplistic, naïve diversification of currency factors. We find that this superiority is sourced from the extra covariance matrix information added to the currency characteristics and expected mean returns, leading to stronger cross-sectional predictability. We also find that international diversification plays an important role in optimising currency portfolios.

Furthermore, we find that the three MV-optimised factors subsume the abnormal return of the naïve factors. Our asset pricing test results suggest that a four-factor model captures the cross-sectional variation in currency returns significantly better than a model using only three naïve factors, providing a novel benchmark for FX investors. The improvement is economically meaningful, as the optimised factors isolate the priced component of currency risk while filtering out return variation that does not command a risk premium.

Because FX investors are primarily concerned with systematic rather than transitory currency fluctuations (Lustig et al., 2011; Atanasov and Nitschka, 2015), the optimised factors have direct relevance for practice. They provide a disciplined basis for constructing currency overlays and dynamic hedges that deliver more stable risk-adjusted performance than naïve carry, momentum, or value portfolios, and offer a sharper proxy for the priced component of currency risk in asset pricing and macro-financial analysis. The optimisation-based currency factors are not merely statistical refinements of existing strategies; they are economically informative benchmarks that improve allocation, hedging, and pricing in international financial markets.

The economic value of the MV-optimised factors is notable, indicating that the profitability of currency factors, when incorporating portfolio optimisation, is substantially improved. Our findings provide valuable insights for investors by presenting a compelling methodology for portfolio allocation decisions. Future research direction can be to examine whether the optimised currency factors are related to the performance of institutional currency investment strategies, such as currency hedge funds (Nucera and Valente, 2013).

CRedit authorship contribution statement

Minyou Fan: Writing – review & editing, Writing – original draft, Validation, Software, Methodology, Formal analysis, Data curation. **Fearghal Kearney:** Writing – review & editing, Writing – original draft, Supervision. **Youwei Li:** Writing – review & editing, Supervision. **Jiadong Liu:** Writing – review & editing, Writing – original draft, Supervision, Project administration.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Appendix A. Summary statistics

This section details our data sample and summary statistics. For data cleaning purposes, we follow Lustig et al. (2011) and Menkhoff et al. (2012a) in identifying episodes of large failures of CIP conditions and liquidity issues in the historical data. That is, we filter out the following observations from our sample: *South African Rand* from 1995 to 2001; *Malaysian Ringgit* from August 1998 to June 2005; *Indonesian Rupiah* from 1999 to 2007; *Hungarian Forint* in 2000; *Philippine Peso* in 1999; *Taiwan Dollar* in 1999;

Thai Baht in 1999; *Egyptian Pound* from December 2011 to March 2017. In addition, we remove *Ukrainian Hryvnia* from June 2015 onwards, because the forward quotes are missing.

Table A.1 presents some descriptive statistics of our FX market data. Most currencies yield positive annualised returns over the sample period, with the *Slovak Koruna* reporting the highest annualised return at 13.37% per annum. Only 15 out of 48 currencies produce negative returns, with *Finnish Markka* displaying the lowest average return, at -7.06% per annum. The annualised standard deviation ranges from 0.003 (*Saudi Arabian Riyal*) to 0.241 (*Indonesian Rupiah*). Finally, the *Egyptian Pound* illustrates the highest Sharpe ratio, at 1.178 per annum, while the *Finnish Markka* shows the lowest value, at -0.691 per annum.

Table A.1
Summary statistics.

Regions/Currency	Statistics					Sample period	
	Ann.Mean	Ann.SD	Ann.SR	min	max	Start Date	End date
Australian Dollar	1.65%	0.116	0.142	-15.11%	10.88%	Jan-85	Dec-24
Austrian Schilling	-6.08%	0.101	-0.605	-6.69%	5.66%	Jan-97	Dec-98
Belgian Franc	3.82%	0.125	0.306	-10.34%	7.36%	Nov-83	Dec-98
Brazilian Real	4.10%	0.154	0.267	-15.64%	12.12%	Apr-04	Dec-24
Bulgarian Lev	-1.28%	0.091	-0.140	-9.14%	10.29%	Apr-04	Dec-24
Canadian Dollar	0.27%	0.073	0.038	-11.66%	9.25%	Jan-85	Dec-24
Croatian Kuna	-0.54%	0.094	-0.057	-9.76%	7.82%	Apr-04	Dec-24
Cypriot Pound	4.65%	0.074	0.630	-4.60%	4.42%	Apr-04	Dec-07
Czech Koruna	1.09%	0.121	0.091	-13.11%	10.10%	Jan-97	Dec-24
Danish Krone	1.32%	0.102	0.129	-10.31%	10.12%	Jan-85	Dec-24
Egyptian Pound	7.99%	0.068	1.178	-15.47%	4.83%	Apr-04	Oct-22
Euro	-1.30%	0.094	-0.138	-9.45%	9.89%	Jan-99	Dec-24
Finnish Markka	-7.06%	0.102	-0.691	-6.72%	5.63%	Jan-97	Dec-98
French Franc	4.04%	0.116	0.347	-9.99%	7.93%	Nov-83	Dec-98
German Mark	2.14%	0.120	0.179	-10.45%	7.19%	Nov-83	Dec-98
Greek Drachma	-4.76%	0.111	-0.427	-10.88%	7.19%	Jan-97	Dec-00
Hong Kong Dollar	-0.30%	0.006	-0.475	-1.34%	0.79%	Nov-83	Dec-24
Hungarian Forint	2.11%	0.136	0.155	-18.88%	11.10%	Nov-97	Dec-24
Icelandic Krona	1.40%	0.138	0.102	-26.71%	20.41%	Apr-04	Dec-24
Indian Rupee	1.19%	0.068	0.174	-8.43%	8.16%	Nov-97	Dec-24
Indonesian Rupiah	-2.64%	0.241	-0.109	-59.81%	34.85%	Jan-97	Dec-24
Irish Pound	1.96%	0.081	0.243	-5.98%	5.33%	Nov-93	Dec-98
Israeli New Shekel	0.84%	0.083	0.101	-7.61%	8.15%	Apr-04	Dec-24
Italian Lira	4.13%	0.117	0.354	-14.11%	8.59%	Apr-84	Dec-98
Japanese Yen	-1.47%	0.110	-0.134	-10.29%	16.14%	Nov-83	Dec-24
Kuwaiti Dinar	0.24%	0.021	0.114	-6.23%	2.15%	Jan-97	Dec-24
Malaysian Ringgit	-1.13%	0.080	-0.140	-12.99%	12.57%	Jan-85	Dec-24
Mexican Peso	3.34%	0.115	0.291	-20.38%	12.24%	Jan-97	Dec-24
Dutch Guilder	2.28%	0.120	0.190	-10.48%	7.55%	Nov-83	Dec-98
New Zealand Dollar	3.80%	0.124	0.306	-14.74%	13.55%	Jan-85	Dec-24
Norwegian Krone	1.02%	0.113	0.090	-11.71%	8.64%	Jan-85	Dec-24
Philippine Peso	0.37%	0.075	0.049	-11.99%	9.28%	Jan-97	Dec-24
Polish Zloty	2.12%	0.134	0.159	-16.41%	10.45%	Mar-02	Dec-24
Portuguese Escudo	-5.26%	0.097	-0.543	-6.51%	5.51%	Jan-97	Dec-98
Russian Ruble	2.07%	0.191	0.108	-31.82%	29.18%	Apr-04	Dec-24
Saudi Riyal	0.15%	0.003	0.519	-1.02%	0.52%	Jan-97	Dec-24
Singapore Dollar	0.16%	0.055	0.028	-8.58%	6.08%	Jan-85	Dec-24
Slovak Koruna	13.37%	0.116	1.150	-9.69%	10.55%	Mar-02	Dec-08
Slovenian Tolar	2.45%	0.081	0.302	-4.54%	4.47%	Apr-04	Dec-06
South African Rand	1.71%	0.161	0.106	-19.69%	13.92%	Nov-83	Dec-24
South Korean Won	-0.01%	0.108	-0.001	-13.72%	12.93%	Mar-02	Dec-24
Spanish Peseta	-4.89%	0.101	-0.483	-7.04%	5.70%	Jan-97	Dec-98
Swedish Krona	0.30%	0.111	0.027	-14.49%	11.06%	Jan-85	Dec-24
Swiss Franc	0.12%	0.111	0.011	-14.11%	12.65%	Nov-83	Dec-24
Taiwan Dollar	-2.06%	0.054	-0.379	-8.76%	5.72%	Jan-97	Dec-24
Thai Baht	0.37%	0.099	0.038	-19.96%	16.33%	Jan-97	Dec-24
Ukrainian Hryvnia	-3.43%	0.198	-0.174	-48.39%	12.77%	Apr-04	Jun-15
UK pound	0.89%	0.101	0.088	-13.02%	13.74%	Nov-83	Dec-24

This table reports the summary statistics of currency excess returns. *Ann.Mean* denotes annualised excess return. *Ann.SD* is the annualised standard deviation of excess returns. *Ann.SR* refers to the Sharpe ratio. *max* and *min* represent the highest and lowest single-month excess return, respectively. *Start Date* and *End Date* are the start and end months of the sample period, respectively.

Appendix B. Performance metrics

We employ various performance metrics to evaluate the out-of-sample performance of both the naïve and optimised factors. Initially, we calculate the mean ($\bar{\mu}_p$) and volatility (σ_p) of the excess returns of the out-of-sample (OOS) factors to serve as fundamental profitability and risk metrics.

In addition, we estimate the abnormal profit, α , based on the factor model of Lustig et al. (2011), as:

$$R_t^p = \alpha + R_t^{Dollar} + \epsilon, \quad (\text{B.1})$$

where R^{Dollar} refers to the returns of the dollar factor.

Subsequently, we apply the widely recognised Sharpe ratio by Sharpe (1966) as a risk-adjusted performance measure:

$$SR_p = \frac{\bar{\mu}_p}{\sigma_p}. \quad (\text{B.2})$$

Furthermore, aligning with Platanakis et al. (2021), we incorporate alternative out-of-sample performance metrics. First, the Certainty Equivalent Return (CER) for mean-variance investors is approximated and computed as:

$$CER = \bar{\mu}_p - \frac{\lambda \sigma^2}{2}, \quad (\text{B.3})$$

where λ represents the relative risk aversion parameter, set at 0.25 as per Platanakis et al. (2021).

Next, the Omega ratio by Keating and Shadwick (2002), with a target return of zero, is computed as the average gain to the average loss ratio:

$$Omega = \frac{\sum_{t=1}^{\tau} \max(0, R_{p,t})}{\sum_{t=1}^T \max(0, -R_{p,t})}, \quad (\text{B.4})$$

where τ signifies the sample size of the OOS observations, T represents the total number of observations, and $R_{p,t}$ denotes the OOS portfolio return at time t . The Omega ratio has the advantage of not relying on assumptions about the distribution of portfolio returns.

The Dowd ratio, introduced by Prigent (2007), is the out-of-sample mean excess portfolio return divided by the portfolio value-at-risk (VaR), which is measured as:

$$Dowd = \frac{\bar{\mu}_p}{VaR_{0.95}}, \quad (\text{B.5})$$

with the VaR calculated at the 95% confidence level over the OOS period.

Appendix C. Abnormal returns based on the multi-factor model

Table C.1 summarises the correlations and the t -statistics of the asset pricing model for each currency over the entire sample period. Panel A contains the regression results of developed currencies, as defined in Section 4, and Panel B lists the results of emerging currencies.

We find that the average alpha (%) in the developed sample, -0.009 , is higher than in the emerging sample, -0.015 . Two alphas in Panel A are significantly higher than zero at the 10% level, and four currencies are significantly greater than zero at the 10% level in Panel B. The standard deviation of α_k in the emerging sample exceeds that of the developed sample, which supports our findings in Section 4. Furthermore, the regression results further show that σ_ϵ in the emerging sample, 0.023 , is higher than in the developed sample, 0.016 .

Table C.1
Currency abnormal profit and idiosyncratic volatility.

	α_k	$t(\alpha_k)$	β_{DXY}	$t(\beta_{DXY})$	σ_ϵ
Panel A: Developed					
Australian Dollar	−0.113	−0.895	0.623	5.520	0.028
Austrian Schilling	−0.135	−0.720	0.985	10.064	0.012
Belgian Franc	0.110	1.232	1.140	36.784	0.012
Canadian Dollar	−0.094	−1.269	0.364	5.914	0.018
Cyprus Pound	0.352***	4.122	0.567	5.121	0.006
Czech Koruna	0.110	1.091	1.307	29.247	0.019
Danish Krone	0.041	0.893	1.145	53.833	0.009
Euro	−0.062	−1.682	1.159	47.947	0.008
Finnish Markka	−0.218	−1.000	1.007	11.117	0.013
French Franc	0.147**	2.261	1.129	39.727	0.011
German Mark	0.016	0.240	1.171	43.701	0.011

(continued on next page)

Table C.1 (continued)

	α_k	$t(\alpha_k)$	β_{DXY}	$t(\beta_{DXY})$	σ_e
Greek Drachma	0.140	0.817	1.050	7.660	0.018
Hong Kong Pound	−0.024**	−2.258	0.012	2.999	0.002
Irish Punt	−0.012	−0.052	0.804	7.806	0.033
Israeli Shekel	0.018	0.132	0.991	13.781	0.013
Italian Lira	0.094	0.815	0.595	10.959	0.020
Icelandic Krona	0.008	0.061	1.033	20.706	0.017
Japanese Yen	−0.005	−0.039	0.714	11.116	0.025
Netherlands Guilder	0.027	0.400	1.177	43.648	0.011
New Zealand Dollar	0.072	0.571	0.791	7.402	0.028
Norwegian Krone	−0.074	−0.919	1.066	23.998	0.018
Portuguese Escudo	−0.079	−0.413	0.945	9.859	0.012
Singapore Dollar	−0.061	−1.081	0.432	12.657	0.011
South Korean Won	−0.175	−1.498	0.875	9.225	0.022
Spanish Peseta	−0.031	−0.199	0.997	8.331	0.012
Swedish Krona	−0.119	−1.668	1.104	27.041	0.017
Swiss Franc	0.010	0.154	1.120	26.998	0.017
Taiwan Dollar	−0.209***	−2.755	0.383	10.284	0.013
UK Pound	−0.007	−0.078	0.904	19.150	0.019
Average	−0.009	−0.129	0.882	19.400	0.016
SD	0.120	1.385	0.307	14.841	0.007
Panel B: Emerging					
Brazilian Real	−0.080	−0.425	0.889	7.805	0.029
Bulgarian Lev	−0.011	−0.290	1.138	39.405	0.008
Croatian Kuna	0.036	0.702	1.143	44.620	0.010
Egyptian Pound	0.689***	3.965	0.112	2.418	0.019
Hungarian Forint	0.001	0.007	1.406	20.360	0.022
Indian Rupee	−0.068	−0.768	0.384	8.947	0.016
Indonesian Rupiah	−0.343	−0.637	0.857	3.312	0.060
Kuwaiti Dinar	0.024	0.918	0.173	9.326	0.005
Malaysian Ringgit	−0.176	−1.335	0.409	7.944	0.020
Mexican Peso	−0.031	−0.186	0.452	5.781	0.026
Philippine Peso	−0.090	−0.711	0.337	6.996	0.018
Polish Zloty	−0.054	−0.385	1.366	19.460	0.021
Russian Ruble	−0.301	−1.259	0.602	3.400	0.045
Saudi Riyal	0.011**	2.456	0.002	1.537	0.001
Slovak Koruna	0.561***	3.034	1.076	6.683	0.015
Slovenian Tolar	0.350***	4.147	0.625	3.827	0.005
South African Rand	−0.273	−1.478	0.849	8.047	0.037
Thai Baht	−0.076	−0.440	0.557	6.881	0.025
Ukrainian Hryvnia	−0.457	−0.955	0.143	0.923	0.051
Average	−0.015	0.335	0.659	10.930	0.023
SD	0.277	1.716	0.420	11.795	0.016

This table reports the regression results for each currency based on Eq. (16). α_k and β represent the intercept and slope coefficients of the dollar index, and $t(\alpha_k)$ and $t(\beta_{DXY})$ report the t -values measured by Heteroskedasticity- and autocorrelation-consistent (HAC) standard errors of Newey and West (1987). σ_e refers to the standard deviation of the noise term ϵ_t in Eq. (16), which is also known as idiosyncratic volatility. ‘*’, ‘**’, ‘***’ represent that the t -values of α_k are statistically significant at the 10%, 5% and 1% levels, respectively.

Data availability

Data will be made available on request.

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